Marine Microbial Biodiversity

Lecture 05

Microbial Metabolism Pt.3

Matteo Selci, PhD

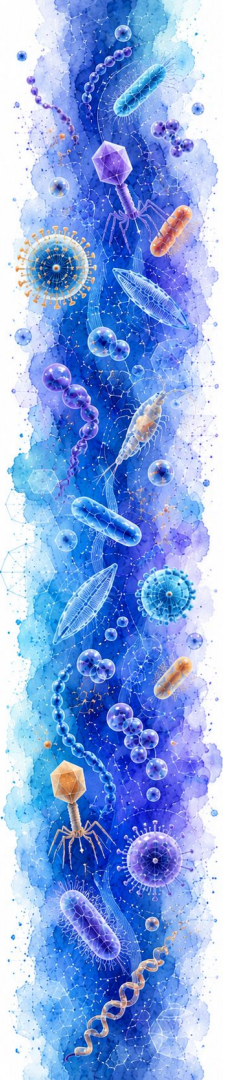
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Chemotrophs

Chemotrophs are organisms that obtain energy through a chemical process called chemosynthesis, rather than by photosynthesis. In chemotrophs, **energy is obtained by chemicals**, organic or inorganic, throughout redox reactions that transfer electrons from an initial electron donor to a final electron acceptor.

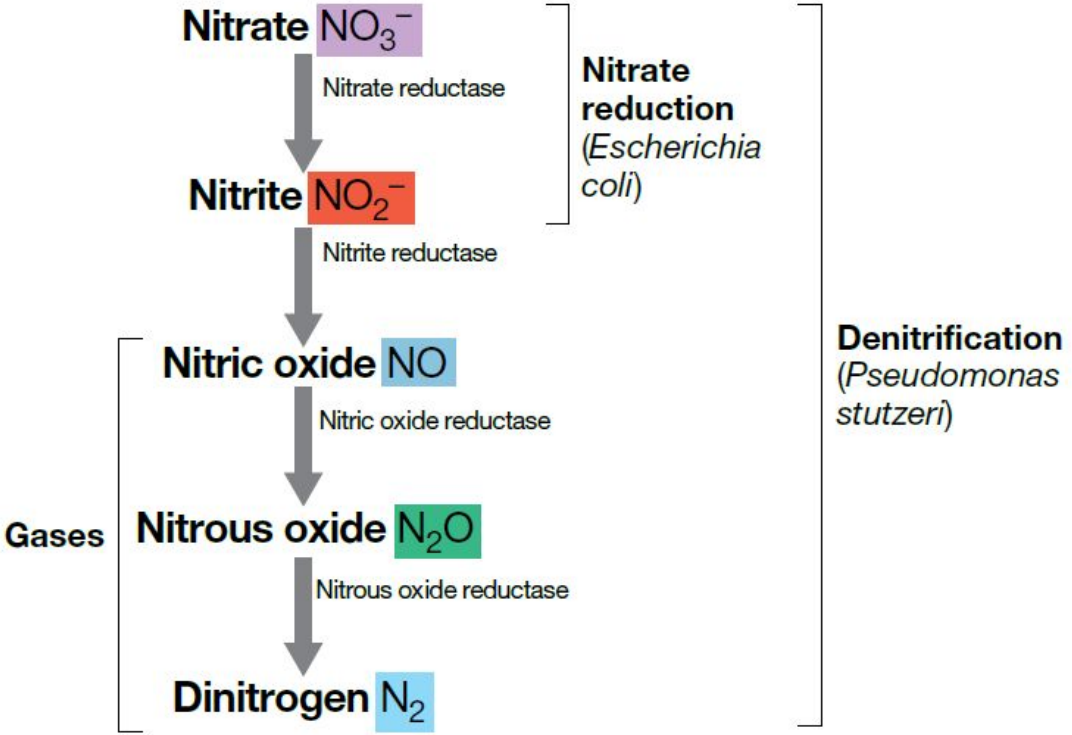
Chemoorganotrophy is a term used to denote the **oxidation of organic chemicals** to yield energy. Chemoorganotrophs are by definition also heterotrophs. They perform catabolism of organic compounds through glycolysis, which could proceed in presence of O_2 with the citric acid cycle and the oxidative phosphorylation, a pathway known as **aerobic respiration**. They use organic compounds both as carbon source and electron donor, and oxygen as terminal electron acceptor.

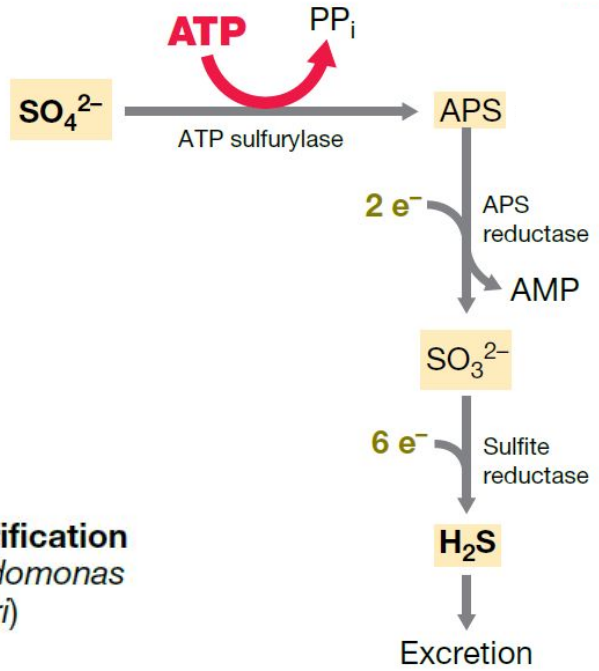
Under **anoxic** conditions, **inorganic electron acceptors** other than O_2 support respiration in a wide variety of microbes in the process called **anaerobic respiration**: ferric iron (Fe^{3+}), nitrate (NO_3^-), and nitrite (NO_2^-), sulfate (SO_4^{2-}), elemental sulfur (S^0), and carbon dioxide (CO_2) among others.

As a third option, **fermentation** is a form of **anaerobic catabolism** in which organic compounds both donate and accept electrons, and redox balance is achieved without the need for external electron acceptors, generating ATP as a result of substrate-level phosphorylation, and not by harnessing an ion motive force.

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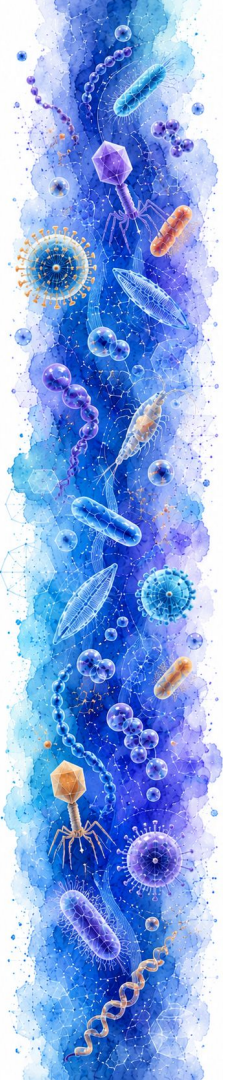
Anaerobic Respiration





Dissimilative **sulfate reduction**

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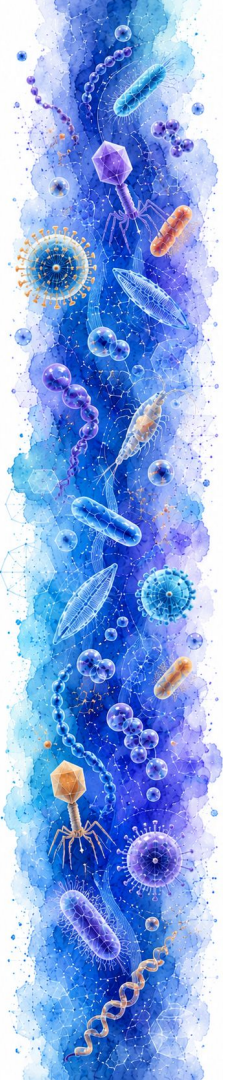
Chemotrophs

Chemotrophs are organisms that obtain energy through a chemical process called chemosynthesis, rather than by photosynthesis. In chemotrophs, **energy is obtained by chemicals**, organic or inorganic, throughout redox reactions that transfer electrons from an initial electron donor to a final electron acceptor.

Chemolithotrophy is a term used to denote the **oxidation of inorganic chemicals** to yield energy. The energetics of chemolithotrophy depend on the free-energy yield of coupled redox reactions between an electron donor and electron acceptor. They can be both autotrophs and heterotrophs, but most are **autotrophs**. They synthesize new organic matter from inorganic carbon (CO_2), therefore known as *"primary producers"*, mostly throughout the **Calvin cycle**, the major biochemical pathway by which autotrophic organisms incorporate CO_2 into cell material (CO_2 fixation).

Chemolithotrophs can tap into many natural sources of **inorganic electron donors**, including geological, biological, and anthropogenic. Electron donors with sufficient **reducing power** to drive ATP synthesis include elemental sulfur (S^0), hydrogen sulfide (H_2S), hydrogen (H_2), ferrous iron (Fe^{2+}), ammonia (NH_3), nitrite (NO_2^-), methane (CH_4), among others.

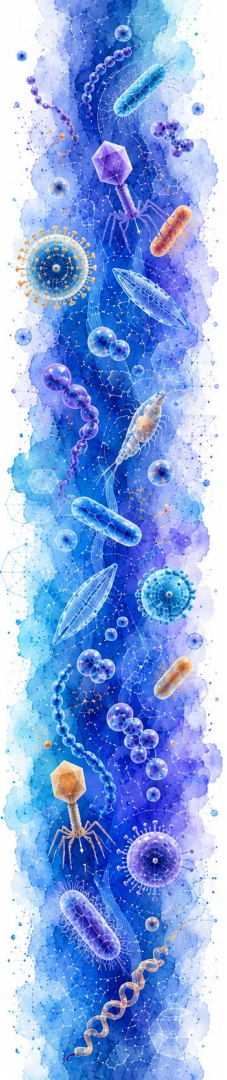
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Chemolithotrophy

Electron donor	*Chemolithotrophic reaction*	*Group of chemolithotrophs*
Hydrogen[b]	$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$	Hydrogen bacteria
Sulfide[b]	$HS^- + H^+ + \frac{1}{2}O_2 \rightarrow S^0 + H_2O$	Sulfur bacteria
Sulfur[b]	$S^0 + 1\frac{1}{2}O_2 + H_2O \rightarrow SO_4^{2-} + 2\,H^+$	Sulfur bacteria
Ammonium[c]	$NH_4^+ + 1\frac{1}{2}O_2 \rightarrow NO_2^- + 2\,H^+ + H_2O$	Nitrifying bacteria
Nitrite[b]	$NO_2^- + \frac{1}{2}O_2 \rightarrow NO_3^-$	Nitrifying bacteria
Ferrous iron[b]	$Fe^{2+} + H^+ + \frac{1}{4}O_2 \rightarrow Fe^{3+} + \frac{1}{2}H_2O$	Iron bacteria

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Types of Chemolithotrophy

e^- donors (reduced forms)

H_2, NH_4^+, H_2S, $S^-_{(n)}$, S^0, $S_2O_3^{2-}$, S^{2-}, CH_4, CO, Fe^{2+}, As^{3+}, etc…

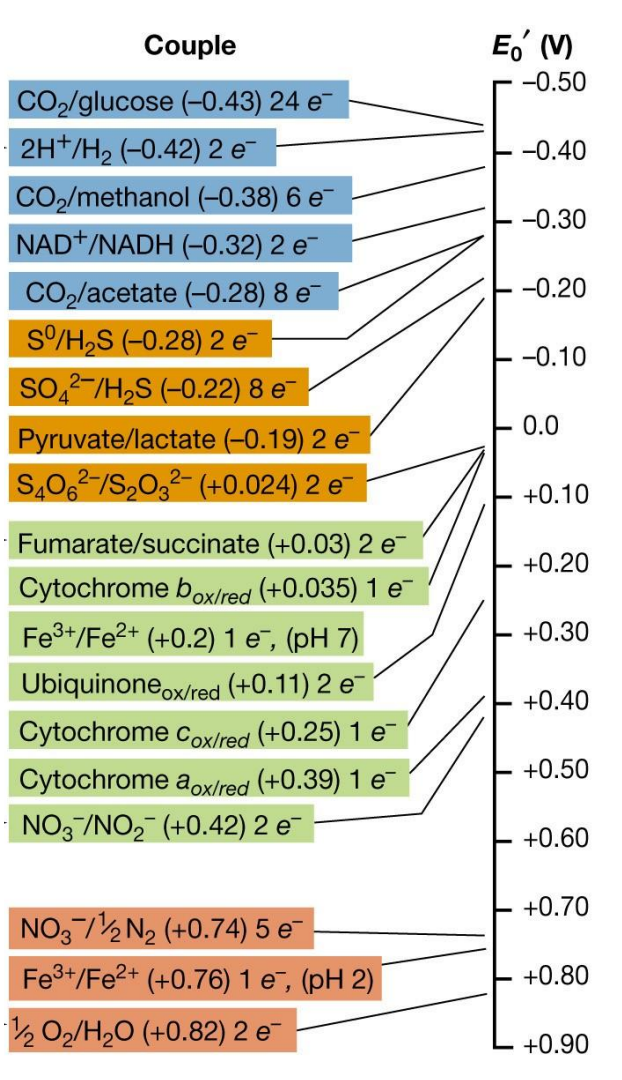
e^- acceptors (oxidized forms)

O_2, NO_3^-, NO_2^-, N_2O, NO, SO_4^{2-}, SO_3^{2-}, S^0, CO_2, Fe^{3+}, Mn^{2+}, SeO_4^{2-}, AsO_4^{3-}, UO_3^{2-}, TeO_4^{2-}, etc…

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The Redox Tower

In a redox tower, the substances are listed according to their reduction potential (E°), which measures their tendency to gain or lose electrons in redox reactions. Each substance in the tower exists as a **redox couple**, represented as the oxidized/reduced form (e.g., SO_4^{2-}/H_2S). The **top** of the tower contains the **strongest electron donors** (those that are in their reduced form and are easily oxidized). These molecules have **negative reduction potentials**, meaning they have a **strong tendency to give electrons**. At the bottom of the tower are the strongest electron acceptors (those that are in their oxidized form and are easily reduced). These have **positive reduction potentials**, meaning they have a **strong tendency to accept electrons**. The **larger** the **difference** in **reduction potential** between the donor and acceptor, the **more energy** is released when electrons are transferred from the donor to the acceptor. For example, when electrons are transferred from a strong donor (like glucose) to a strong acceptor at the bottom (like O_2), a large amount of free energy is released, which can be harnessed by cells to do work, such as generating ATP.



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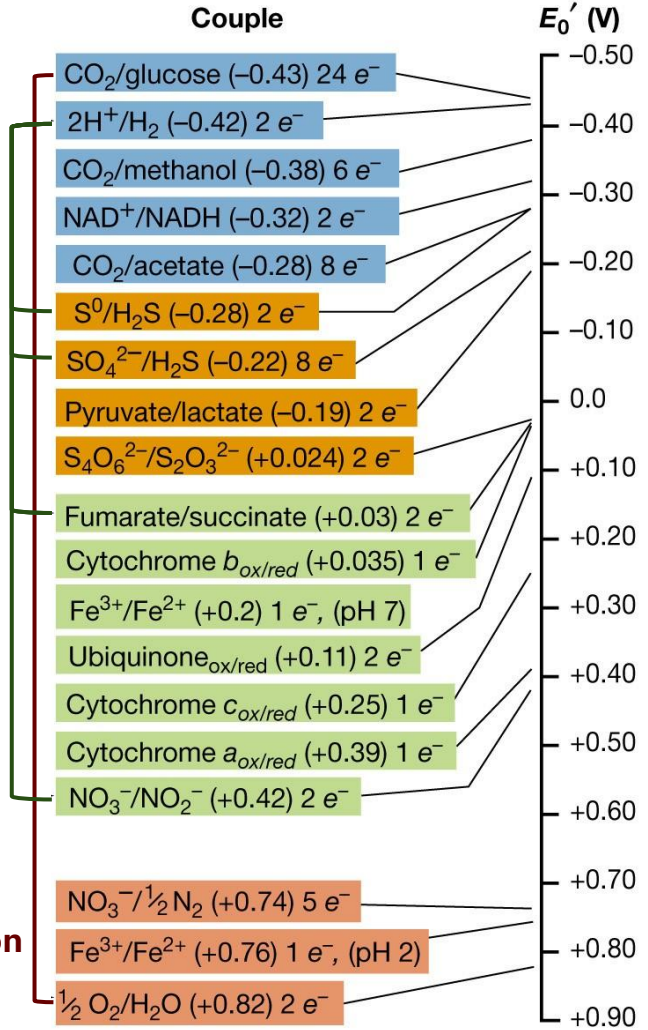
The Redox Tower

Hydrogenotrophic pathways include:
Nitrate Reduction
Sulfate Reduction
Methanogenesis
Acetogenesis
Iron and Manganese Reduction
Fumarate / Organic Reduction

Anaerobic Respiration

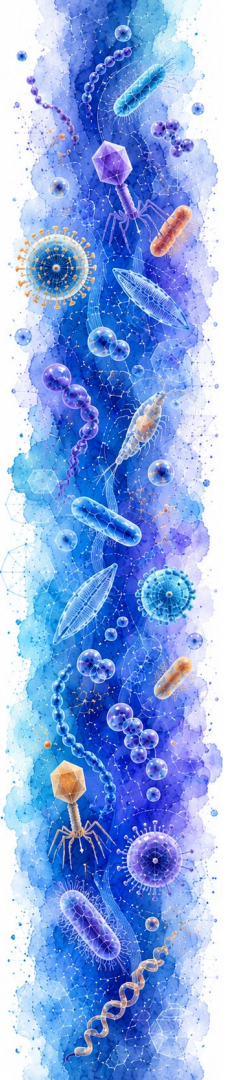
Chemoorganoheterotrophy pathway:
Glycolysis
Krebs cycle
Oxidative Phosphorylation

Aerobic Respiration





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Redox In Chemolithoautotrophy

Reaction

$2NH_4^+ + 1.5O_2 \rightarrow N_2 + 2H^+ + 3H_2O$
$NH_4^+ + 1.5O_2 \rightarrow NO_2^- + 2H^+ + H_2O$
$NH_4^+ + 2O_2 \rightarrow NO_3^- + 2H^+ + H_2O$
$N_2 + 1.5O_2 + H_2O \rightarrow 2NO_2^- + 2H^+$
$N_2 + 2.5O_2 + H_2O \rightarrow 2NO_3^- + 2H^+$
$NO_2^- + 0.5O_2 \rightarrow NO_3^-$
$NO_3^- + H_2 \rightarrow NO_2^- + H_2O$
$2NO_3^- + 5H_2 + 2H^+ \rightarrow N_2 + 6H_2O$
$NO_3^- + 4H_2 + 2H^+ \rightarrow NH_4^+ + 3H_2O$
$2NO_2^- + 3H_2 + 2H^+ \rightarrow N_2 + 4H_2O$
$NO_2^- + 3H_2 + 2H^+ \rightarrow NH_4^+ + 2H_2O$
$N_2 + 3H_2 + 2H^+ \rightarrow 2NH_4^+$
$4N_2 + 2H^+ + 9H_2O \rightarrow 3NO_3^- + 5NH_4^+$
$N_2 + 2H_2O \rightarrow NO_2^- + NH_4^+$
$4NO_2^- + 2H^+ + H_2O \rightarrow 3NO_3^- + NH_4^+$
$5NO_2^- + 2H^+ \rightarrow 3NO_3^- + N_2 + H_2O$

$H_2S + 0.5O_2 \rightarrow S^\circ + H_2O$
$H_2S + 2O_2 \rightarrow SO_4^{2-} + 2H^+$
$S^\circ + 1.5O_2 + H_2O \rightarrow SO_4^{2-} + 2H^+$
$SO_4^{2-} + 3H_2 + 2H^+ \rightarrow S^\circ + 4H_2O$
$SO_4^{2-} + 4H_2 + 2H^+ \rightarrow H_2S + 4H_2O$
$S^\circ + H_2 \rightarrow H_2S$
$4S^\circ + 4H_2O \rightarrow SO_4^{2-} + 3H_2S + 2H^+$

$CH_4 + 1.5O_2 \rightarrow CO + 2H_2O$
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
$CO + 0.5O_2 \rightarrow CO_2$
$CO_2 + H_2 \rightarrow CO + H_2O$
$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$
$CO + 3H_2 \rightarrow CH_4 + H_2O$
$4CO + 2H_2O \rightarrow 3CO_2 + CH_4$

$2NH_4^+ + 3Fe_3O_4 + 16H^+ \rightarrow N_2 + 9Fe^{2+} + 12H_2O$
$NH_4^+ + 3Fe_3O_4 + 16H^+ \rightarrow NO_2^- + 9Fe^{2+} + 10H_2O$
$NH_4^+ + 4Fe_3O_4 + 22H^+ \rightarrow NO_3^- + 12Fe^{2+} + 13H_2O$
$N_2 + 3Fe_3O_4 + 16H^+ \rightarrow 2NO_2^- + 9Fe^{2+} + 8H_2O$
$N_2 + 5Fe_3O_4 + 28H^+ \rightarrow 2NO_3^- + 15Fe^{2+} + 14H_2O$
$NO_2^- + Fe_3O_4 + 6H^+ \rightarrow NO_3^- + 3Fe^{2+} + 3H_2O$

$3Fe^{2+} + 0.5O_2 + 3H_2O \rightarrow Fe_3O_4 + 6H^+$
$Fe_3O_4 + H_2 + 6H^+ \rightarrow 3Fe^{2+} + 4H_2O$

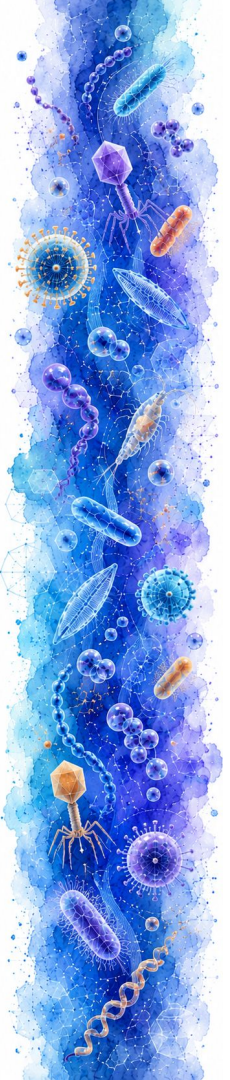
$2NH_4^+ + SO_4^{2-} \rightarrow N_2 + S^\circ + 4H_2O$
$8NH_4^+ + 3SO_4^{2-} \rightarrow 4N_2 + 3H_2S + 2H^+ + 12H_2O$
$NH_4^+ + SO_4^{2-} \rightarrow NO_2^- + S^\circ + 2H_2O$
$4NH_4^+ + 3SO_4^{2-} \rightarrow 4NO_2^- + 3H_2S + 2H^+ + 4H_2O$
$3NH_4^+ + 4SO_4^{2-} + 2H^+ \rightarrow 3NO_3^- + 4S^\circ + 7H_2O$
$NH_4^+ + SO_4^{2-} \rightarrow NO_3^- + H_2S + H_2O$
$2NH_4^+ + 3S^\circ \rightarrow N_2 + 3H_2S + 2H^+$
$NH_4^+ + 3S^\circ + 2H_2O \rightarrow NO_2^- + 3H_2S + 2H^+$
$NH_4^+ + 4S^\circ + 3H_2O \rightarrow NO_3^- + 4H_2S + 2H^+$
$N_2 + SO_4^{2-} \rightarrow 2NO_2^- + S^\circ$
$4N_2 + 3SO_4^{2-} + 4H_2O \rightarrow 8NO_2^- + 3H_2S + 2H^+$
$3N_2 + 5SO_4^{2-} + 4H^+ \rightarrow 6NO_3^- + 5S^\circ + 2H_2O$
$4N_2 + 5SO_4^{2-} + 2H^+ + 4H_2O \rightarrow 8NO_3^- + 5H_2S$
$N_2 + 3S^\circ + 4H_2O \rightarrow 2NO_2^- + 3H_2S + 2H^+$
$N_2 + 5S^\circ + 6H_2O \rightarrow 2NO_3^- + 5H_2S + 2H^+$
$3NO_2^- + SO_4^{2-} + 2H^+ \rightarrow 3NO_3^- + S^\circ + H_2O$
$4NO_2^- + SO_4^{2-} + 2H^+ \rightarrow 4NO_3^- + H_2S$
$NO_2^- + S^\circ + H_2O \rightarrow NO_3^- + H_2S$

$2NH_4^+ + 3CO_2 \rightarrow N_2 + 3CO + 2H^+ + 3H_2O$
$8NH_4^+ + 3CO_2 \rightarrow 4N_2 + 3CH_4 + 8H^+ + 6H_2O$
$NH_4^+ + 3CO_2 \rightarrow NO_2^- + 3CO + 2H^+ + H_2O$
$4NH_4^+ + 3CO_2 + 2H_2O \rightarrow 4NO_2^- + 3CH_4 + 8H^+$
$NH_4^+ + 4CO_2 \rightarrow NO_3^- + 4CO + 2H^+ + H_2O$
$NH_4^+ + CO_2 + H_2O \rightarrow NO_3^- + CH_4 + 2H^+$
$2NH_4^+ + CO \rightarrow N_2 + CH_4 + 2H^+ + H_2O$
$NH_4^+ + CO + H_2O \rightarrow NO_2^- + CH_4 + 2H^+$
$3NH_4^+ + 4CO + 5H_2O \rightarrow 3NO_3^- + 4CH_4 + 6H^+$
$N_2 + 3CO_2 + H_2O \rightarrow 2NO_2^- + 3CO + 2H^+$
$4N_2 + 3CO_2 + 10H_2O \rightarrow 8NO_2^- + 3CH_4 + 8H^+$
$N_2 + 5CO_2 + H_2O \rightarrow 2NO_3^- + 5CO + 2H^+$
$4N_2 + 5CO_2 + 14H_2O \rightarrow 8NO_3^- + 5CH_4 + 8H^+$
$N_2 + CO + 3H_2O \rightarrow 2NO_2^- + CH_4 + 2H^+$
$3N_2 + 5CO + 13H_2O \rightarrow 6NO_3^- + 5CH_4 + 6H^+$
$NO_2^- + CO_2 \rightarrow NO_3^- + CO$
$4NO_2^- + CO_2 + 2H_2O \rightarrow 4NO_3^- + CH_4$
$3NO_2^- + CO + 2H_2O \rightarrow 3NO_3^- + CH_4$

$H_2S + CO_2 \rightarrow S^\circ + CO + H_2O$
$4H_2S + CO_2 \rightarrow 4S^\circ + CH_4 + 2H_2O$
$H_2S + 4CO_2 \rightarrow SO_4^{2-} + 4CO + 2H^+$
$H_2S + CO_2 + 2H_2O \rightarrow SO_4^{2-} + CH_4 + 2H^+$
$3H_2S + CO \rightarrow 3S^\circ + CH_4 + H_2O$
$3H_2S + 4CO + 8H_2O \rightarrow 3SO_4^{2-} + 4CH_4 + 6H^+$
$S^\circ + 3CO_2 + H_2O \rightarrow SO_4^{2-} + 3CO + 2H^+$
$4S^\circ + 3CO_2 + 10H_2O \rightarrow 4SO_4^{2-} + 3CH_4 + 8H^+$
$S^\circ + CO + 3H_2O \rightarrow SO_4^{2-} + CH_4 + 2H^+$

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10.1046/j.1472-4669.2003.00006.x



Redox In Chemolithoautotrophy

Reaction

$2NH_4^+ + 1.5O_2 \rightarrow N_2 + 2H^+ + 3H_2O$
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$N_2 + 2.5O_2 + H_2O \rightarrow 2NO_3^- + 2H^+$
$NO_2^- + 0.5O_2 \rightarrow NO_3^-$
$NO_3^- + H_2 \rightarrow NO_2^- + H_2O$
$2NO_3^- + 5H_2 + 2H^+ \rightarrow N_2 + 6H_2O$
$NO_3^- + 4H_2 + 2H^+ \rightarrow NH_4^+ + 3H_2O$
$2NO_2^- + 3H_2 + 2H^+ \rightarrow N_2 + 4H_2O$
$NO_2^- + 3H_2 + 2H^+ \rightarrow NH_4^+ + 2H_2O$
$N_2 + 3H_2 + 2H^+ \rightarrow 2NH_4^+$
$4N_2 + 2H^+ + 9H_2O \rightarrow 3NO_3^- + 5NH_4^+$
$N_2 + 2H_2O \rightarrow NO_2^- + NH_4^+$
$4NO_2^- + 2H^+ + H_2O \rightarrow 3NO_3^- + NH_4^+$
$5NO_2^- + 2H^+ \rightarrow 3NO_3^- + N_2 + H_2O$

$H_2S + 0.5O_2 \rightarrow S^\circ + H_2O$
$H_2S + 2O_2 \rightarrow SO_4^{2-} + 2H^+$
$S^\circ + 1.5O_2 + H_2O \rightarrow SO_4^{2-} + 2H^+$
$SO_4^{2-} + 3H_2 + 2H^+ \rightarrow S^\circ + 4H_2O$
$SO_4^{2-} + 4H_2 + 2H^+ \rightarrow H_2S + 4H_2O$
$S^\circ + H_2 \rightarrow H_2S$
$4S^\circ + 4H_2O \rightarrow SO_4^{2-} + 3H_2S + 2H^+$

$CH_4 + 1.5O_2 \rightarrow CO + 2H_2O$
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
$CO + 0.5O_2 \rightarrow CO_2$
$CO_2 + H_2 \rightarrow CO + H_2O$
$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$
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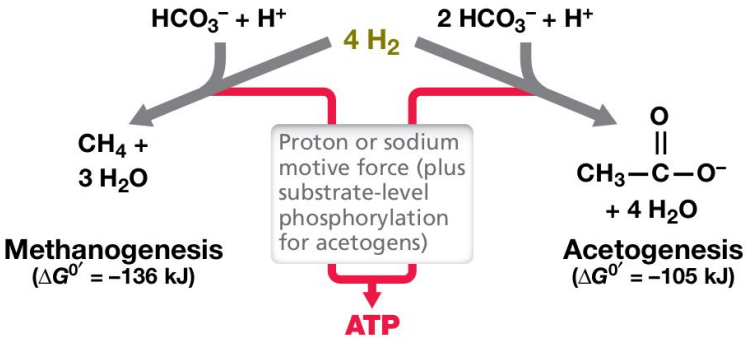
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$S^\circ + 3CO_2 + H_2O \rightarrow SO_4^{2-} + 3CO + 2H^+$
$4S^\circ + 3CO_2 + 10H_2O \rightarrow 4SO_4^{2-} + 3CH_4 + 8H^+$
$S^\circ + CO + 3H_2O \rightarrow SO_4^{2-} + CH_4 + 2H^+$

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10.1046/j.1472-4669.2003.00006.x

Acetogenesis And Methanogenesis

A wide diversity of microbes have evolved metabolic pathways that conserve energy from the reduction of carbon dioxide, so **CO_2 is the electron acceptor**. In this processes, **H_2 is used as electron donor**, which combining with CO_2 forms methane (CH_4) in Methanogenesis, or acetate (CH_3COO^-) in Acetogenesis. These are forms of **anaerobic respiration in which CO_2 is both electron acceptor AND carbon source**. Methanogenesis and is carried out by a strictly anaerobic group of Archaea. Acetogenesis is carried out prevalently by gram positive Bacteria (especially members of the genera Clostridium or Acetobacterium).
Energy conservation in **acetogenesis** results from both substrate-level **phosphorylation** and **proton motive force** (PMF). In **methanogenesis** substrate-level **phosphorylation does not occur**.



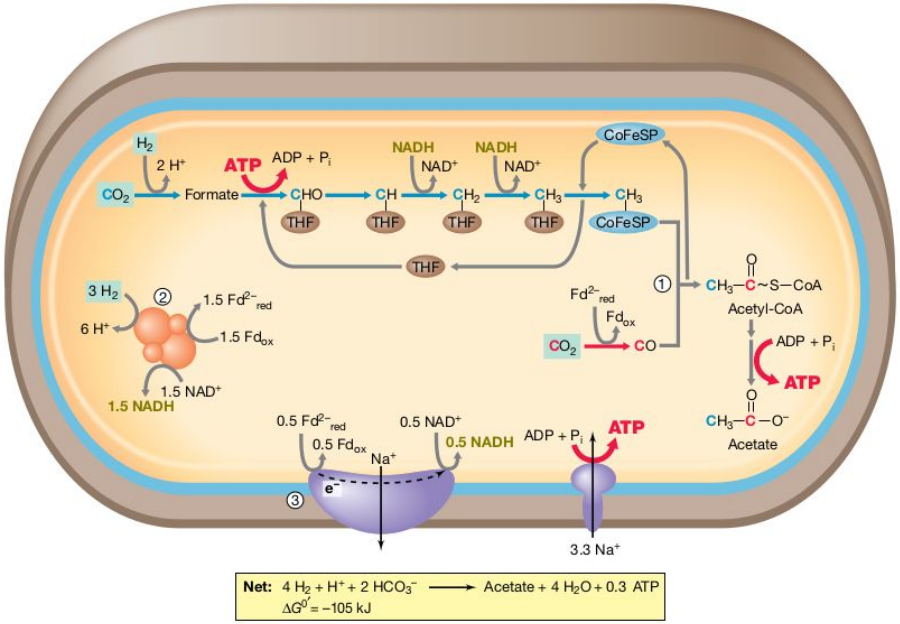
$$4H_2 + H^+ + 2HCO_3^- \rightarrow CH_3COO^- + 4H_2O$$

$$4H_2 + H^+ + HCO_3^- \rightarrow CH_4 + 3H_2O$$

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Acetogenesis

Acetogens are **obligately anaerobic** bacteria that reduce CO_2 to acetate by the **reductive acetyl-CoA or Wood-Ljungdahl pathway**. Some acetogens can grow **chemoorganotrophically**, using **glucose** in glycolysis as electron donor, and the obtained **pyruvate** is oxidized to **acetate** and **CO_2**, which is then used as a terminal electron acceptor in the reductive acetyl-CoA pathway. Others are **chemolithoautotrophs** through the reduction of **CO_2** to **acetate** with **H_2** as electron donor. In *Acetobacterium woodii* the PMF is generated by activity of the Rnf complex (Ferredoxin - NAD^+ Oxidoreductase), sort of electron transport chain involving a reduced ferredoxin (Fd2-red) as electron donor and NAD^+ as electron acceptor, and a Na^+ motive force.



Acetobacterium woodii

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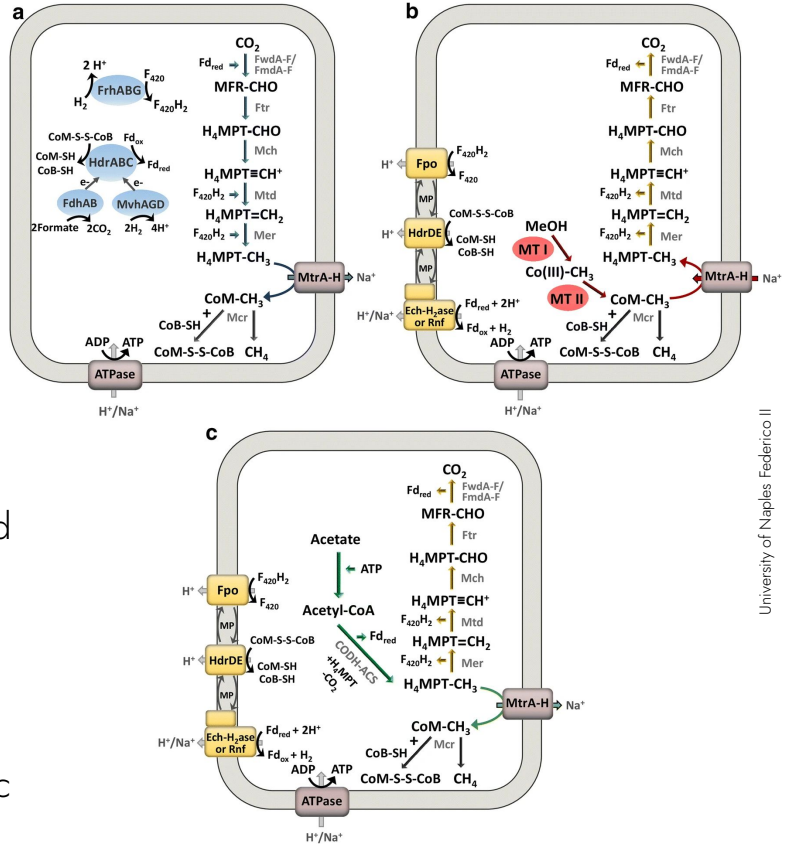
Methanogenesis

Methanogens group of strictly anaerobic Archaea that use a modified Wood–Ljungdahl pathway (**Wolfe cycle**) to fix CO_2 for cell growth via acetyl-coenzyme A (acetyl-CoA) as an intermediate. They are classified into three groups based on electron donors use.

Hydrogenotrophic methanogens oxidize H_2 to reduce CO_2 to CH_4, and represent the predominant source of methanogenesis in deep marine sediments and other anoxic systems.

Methylotrophic methanogens are common in marine and hypersaline, sulfate-rich sediments where they utilize methylated compounds such as trimethylamine, dimethyl sulfate, and methanol.

Aceticlastic methanogens split acetate to form CH_4 and CO_2. They are found in habitats where hydrogenotrophic methanogens reduce H_2 levels low enough to create the conditions needed for high levels of acetate formation by fermenters.



Hydrogenotrophic (a), Methylotrophic (b) and Aceticlastic (c) methanogens

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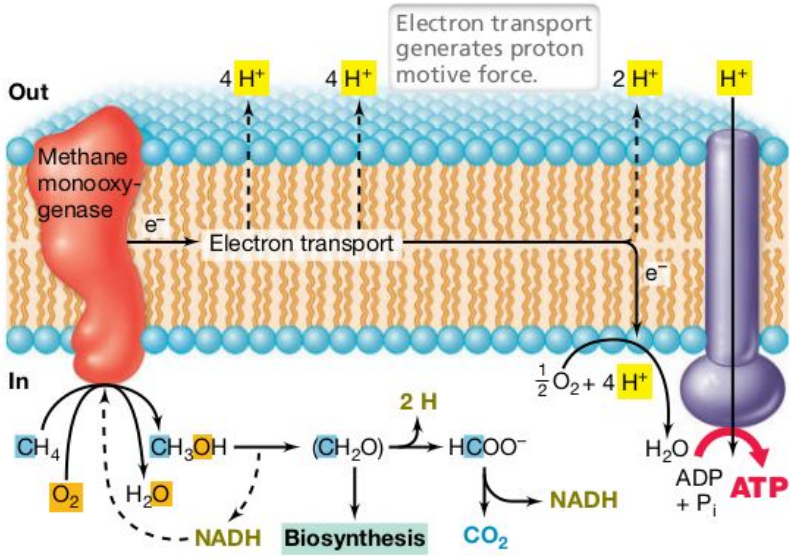
10.1007/s00253-020-10724-7

Methane Oxidation (Methanotrophy)

Methylotrophs are organisms that use **organic compounds** that **lack C-C bonds** as **electron donors and carbon sources**, and in case of methane they are known as methanotrophs.

The aerobic (O_2 as electron acceptor) oxidation of CH_4 has been well studied in the gram-negative bacterium *Methylococcus capsulatus*.

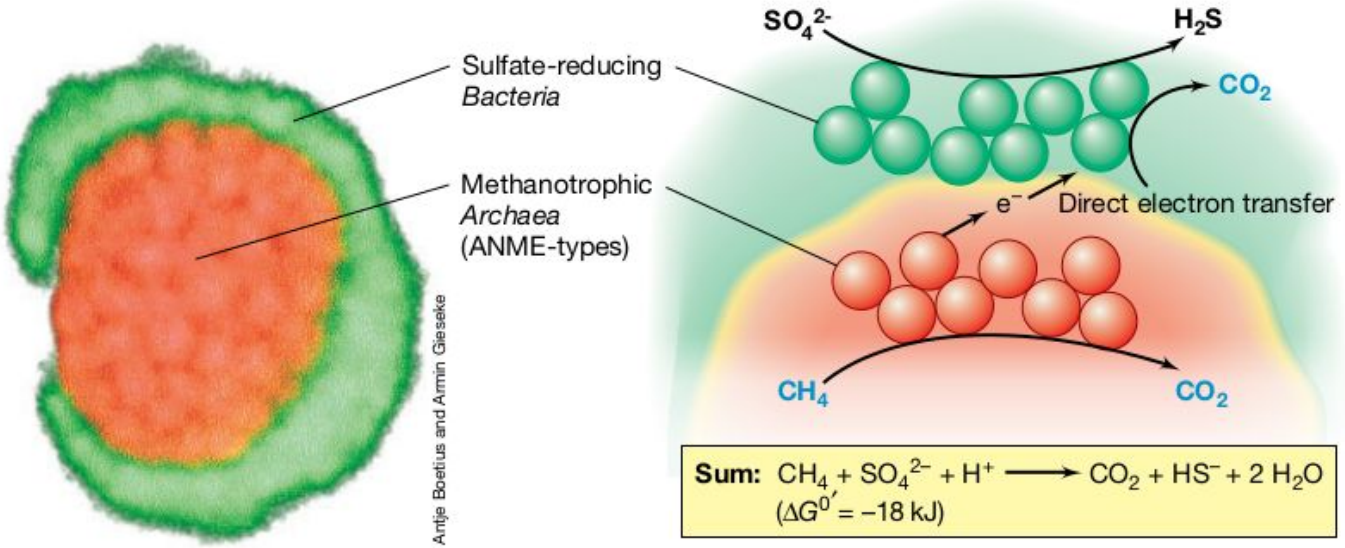
The first step is catalyzed by **methane monooxygenases**, which incorporate one oxygen atom from O_2 into a carbon compound, in this case CH_4, forming methanol (CH_3OH), and the second atom of O is reduced to form H_2O. CH_3OH is oxidized by an **alcohol dehydrogenase**, yielding formaldehyde (CH_2O) which is further oxidized producing CO_2, used to make new cell material, and NADH required for **carbon fixation**.



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Methane Oxidation (Methanotrophy)

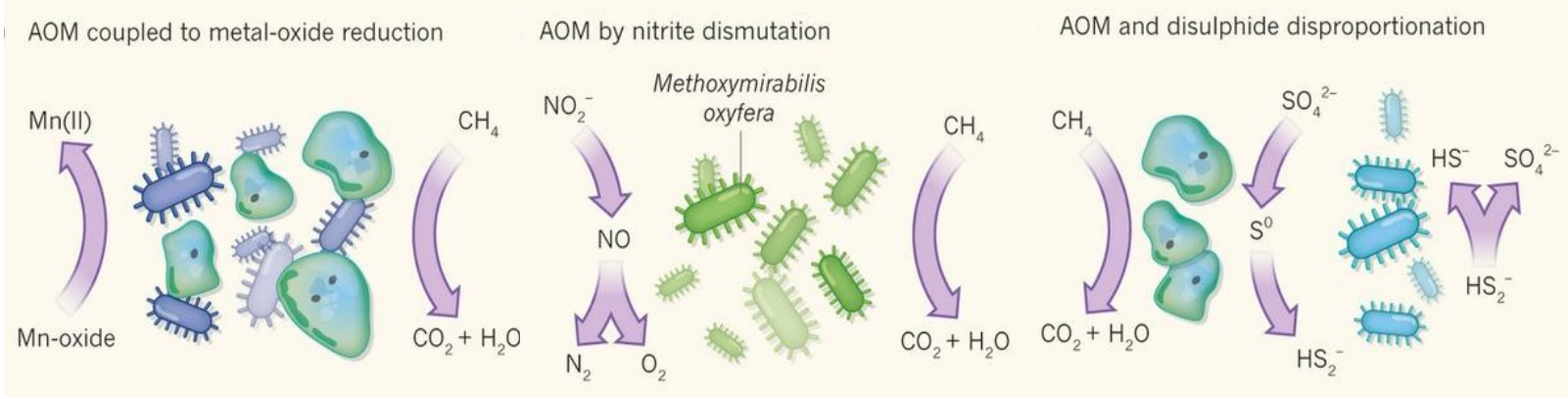
The **anaerobic oxidation of CH_4**, is carried out by a **consortium** of two organisms, a member of a **polyphyletic** group of **Archaea called ANME** (anaerobic methanotroph), which oxidize CH_4 to CO_2 by reverse methanogenesis, thermodynamically unfavorable under anaerobic conditions, plus a **sulfate-reducing bacterium**, which uses electrons from methane oxidation to reduce SO_4^{2-} (electron acceptor) to H_2S. These consortia thrive in anoxic marine sediments and are responsible for oxidizing more than 90% of the methane released from the seafloor (methane hydrates).



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Anaerobic Respiration: Nitrate Reduction and Denitrification

Recently, **three more pathways of anaerobic oxidation of methane** have been reported.
1) The oxidation of CH_4 to CO_2 by ANMEs is coupled to the **reduction of** environmental **metal-oxides**, such as manganese (Mn) or iron (Fe) which are reduced to the +2 oxidation state.
2) The bacterium *Methoxymirabilis oxyfera* reduces **nitrite** (NO_2^-) to **nitric oxide** (NO) and then dismutates NO into **nitrogen** (N_2) and **oxygen** (O_2) as diatomic gases. The bacterium then uses the resulting O_2 to support (intra-aerobic) methane oxidation.
3) Some ANMEs **couple methane oxidation** with **sulphate reduction** to **elemental sulphur** (S^0) and then to **disulphide** (HS^{2-}). The disulphide can be disproportionated by association with sulfur-oxidizing Deltaproteobacteria.

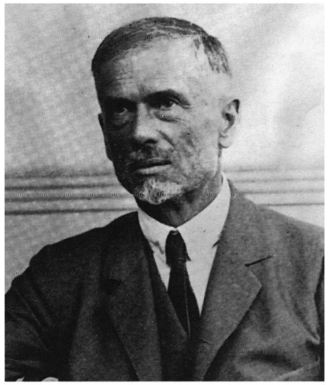


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doi.org/10.1038/nature11749

The History Of Chemolithotrophy

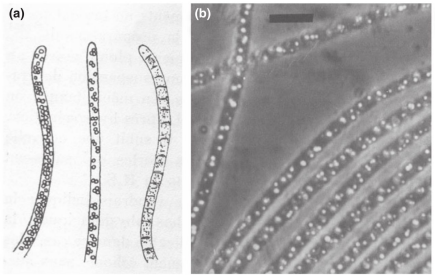
In 1885 Winogradsky entered the laboratory of Anton deBary at the University of Strassburg, to study plant pathology. Winogradsky was assigned to confirm deBary's position that bacteria manifested fixed, stable, and definable properties and could thus indeed be subjected to a botanical style system of taxonomy.
He chose to work on the filamentous bacterium Beggiatoa, known to accumulate sulfur granules.



1856-1953

In his memoirs, in a more informal fashion Winogradsky described the moment that he arrived at this hypothesis. 'Then one day as I was following the Île canal on my way home after a tiring day of chemical work which also involved hydrogen sulfide and sulfur, it suddenly occurred to me that sulfur might be oxidized by *Beggiatoa* to sulfuric acid. I could at once appreciate all the significance and implications of my conjecture, having no doubts that it offered the solution to my problem… The work was humdrum, it dragged on and on sluggishly, and all of a sudden it developed into an interesting result and was finished. All the beating around suddenly made sense, and I matured in my own eyes. Even so, I could not see that my discovery would become an epoch-making discovery, would determine the course of all my future work, and that it would open a new chapter in microbiology and physiology.' (Zavarzin, 1989). What a wonderful description of one of those 'aha!' moments that make scientific research such a unique thrill.

with the disappearance of the sulfur granules (Winogradsky, 1887). It is hard to avoid being thrilled by the next sentence. '…the sulfur in these organisms is the sole respiratory source, and in that sense plays the same role as that of carbohydrate in other organisms.' (Winogradsky, 1887). DeBary referred to his student's discovery as follows 'Sie haben einen neuen Modus vivendi gefunden.' (Zavarzin, 1989). And so indeed he had. It was what later became the concept of autotrophy, coming as a result of his work on the nitrifying bacteria; at this moment, however, he had formulated the concept of chemolithotrophy, the ability of bacteria to obtain energy by the oxidation of reduced inorganic compounds such as H_2S, NH_3, Fe^{2+}, and which led eventually to the entire concept of sulfur and nitrogen cycles in Nature.



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HPLS (2018) 40:11
https://doi.org/10.1007/s40656-017-0175-9

ORIGINAL PAPER

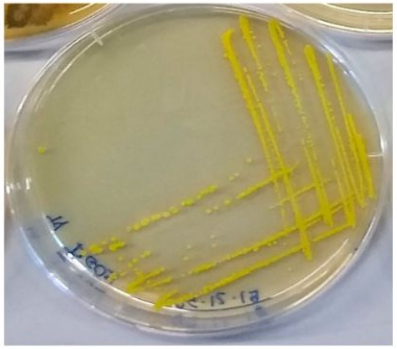
Petri dish versus Winogradsky column: a longue durée perspective on purity and diversity in microbiology, 1880s–1980s

Mathias Grote[1]

In 1885 Winogradsky entered the laboratory of Anton deBary at the University of Strassburg, to study plant pathology. Winogradsky was assigned to confirm deBary's position that bacteria manifested fixed, stable, and definable properties and could thus indeed be subjected to a botanical style system of taxonomy.
He chose to work on the filamentous bacterium Beggiatoa, known to accumulate sulfur granules.



Winogradsky column
Developed by Sergei Winogradsky in 1880s as a way to reproduce complex trophic and environmental interactions among microbes



Petri Plate
Developed by Richard Julius Petri in 1887 as a improvement to Robert Koch plating techniques to isolate pure bacteria

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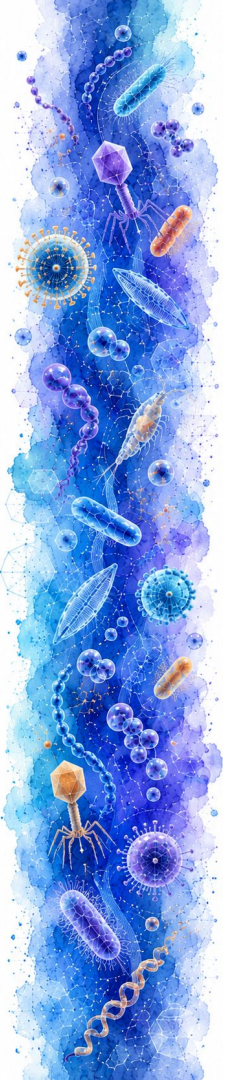


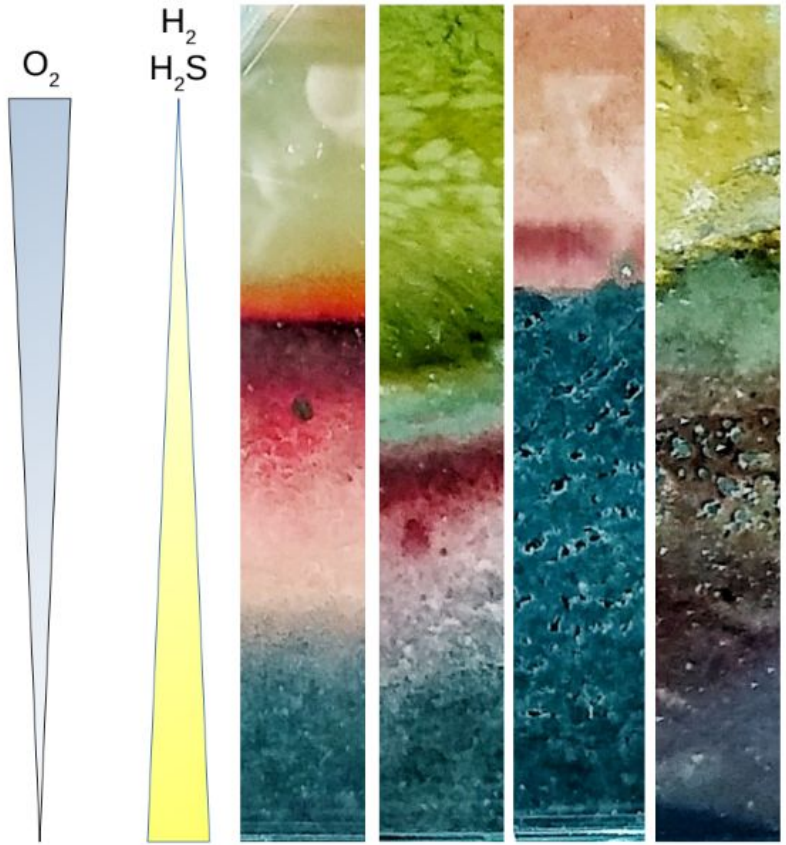
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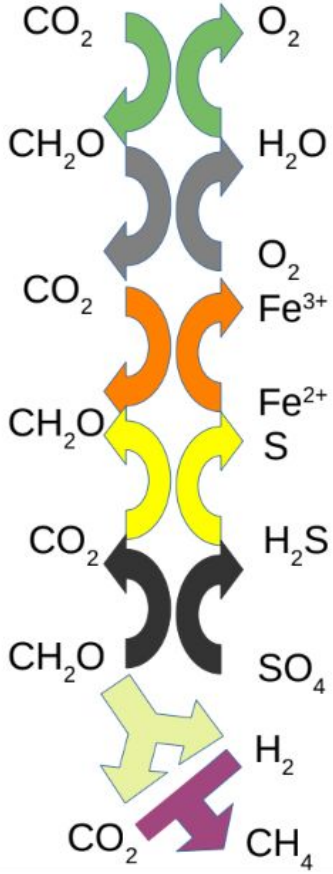


https://joyfulmicrobe.com/winogradsky-column/

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Oxygenic photosynthesis

Heterotrophy

(photo) Iron oxidation

(photo) Sulfur oxidation

Sulfate reduction

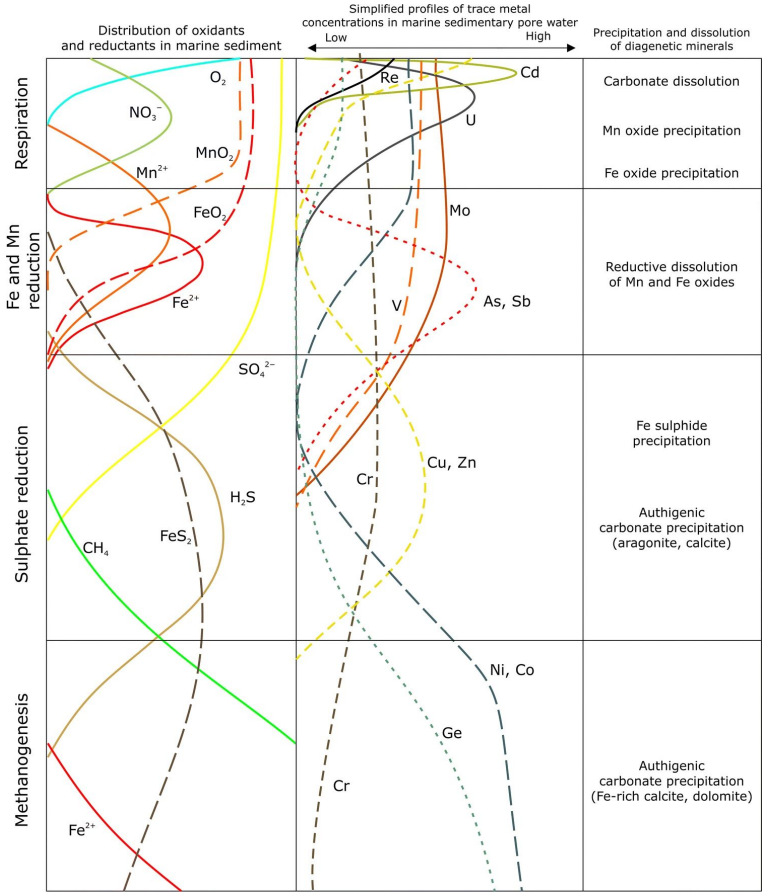
Fermentation

Methanogenesis

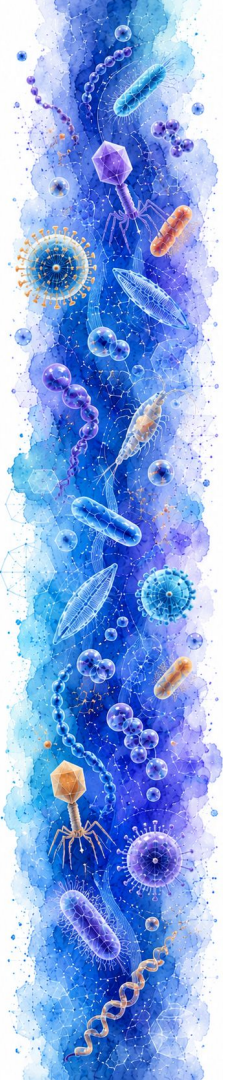
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Redox Zones In Pore Waters Of Marine Sediments

Generally speaking, in any system were nutrient exchange is primarily controlled by diffusion or where mixing is limited, the utilization of the **available electron acceptors** follows a predictable pattern, dictated by their thermodynamic drive. This, together with other more local factors, controls the distribution of different metabolisms at the micro scale.



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Redox Zones In Pore Waters Of Marine Sediments

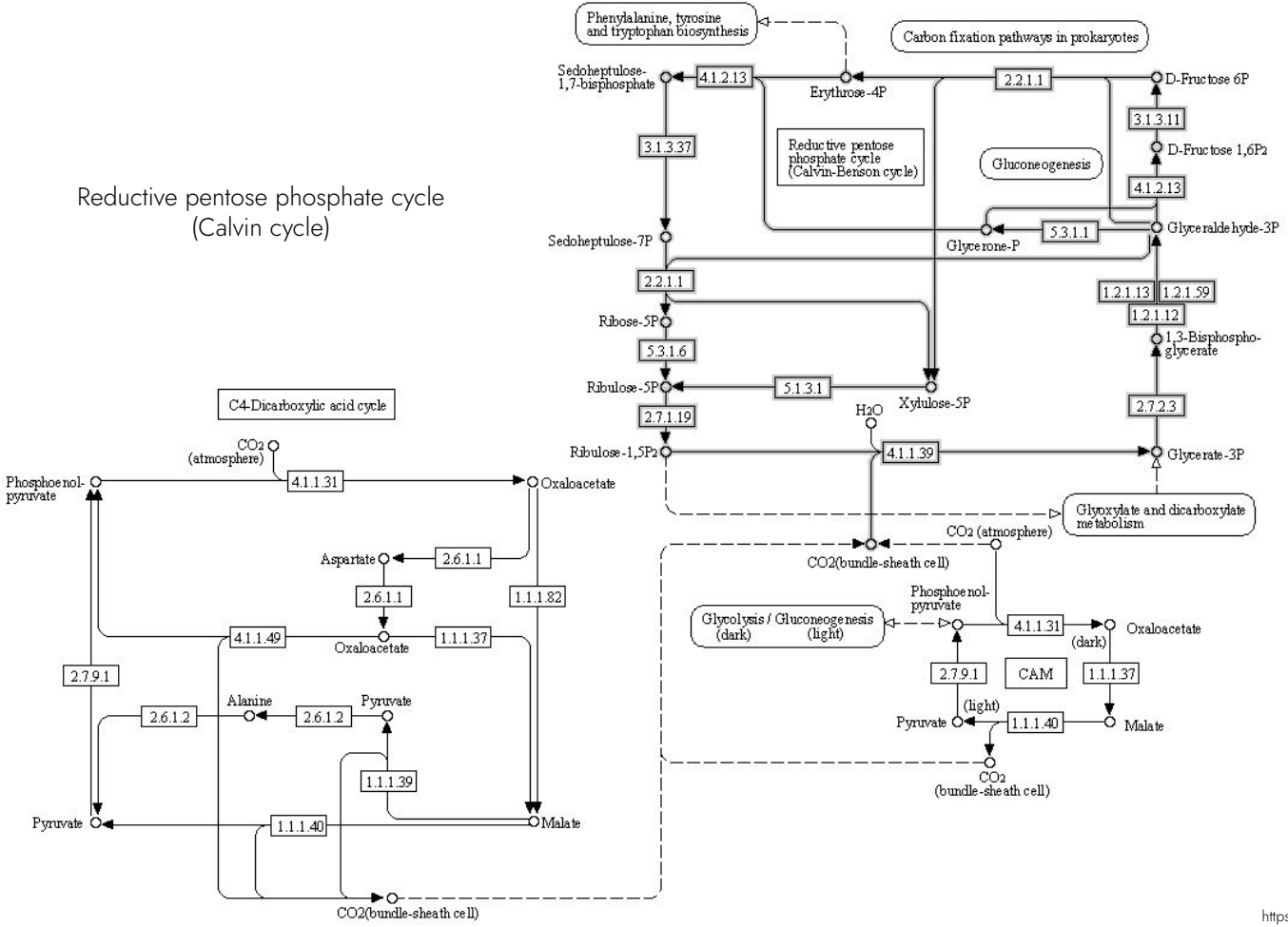
Autotrophy is the process by which an energy-poor and highly oxidized form of carbon, **CO_2**, is reduced to **formate** or **carbon monoxide** (CO) and then assimilated into cell material. However, CO_2 has a negative reduction potential (E0′ = -1,90 V) and its reduction to formate or CO has positive values free Gibbs energy (ΔG0′ = 257.4 kJ mol−1 ; 43.3 kJ mol−1), and are therefore energy-demanding **endergonic** reactions.

In **photoautotrophic** organisms, the **assimilatory** reactions responsible for **CO_2 fixation** are often called the light-independent dark reactions of photosynthesis.
In oxygenic phototrophic **cyanobacteria**, CO_2 is reduced to the level of glyceraldehyde 3-phosphate by the reductive pentose phosphate cycle, or **Calvin-Benson-Bassham cycle**.
Among anoxygenic phototrophs, green sulfur bacteria (*Chlorobi*) fix CO_2 by a reversal of steps in the Krebs cycle, in a pathway therefore known as reductive TCA cycle, or **Arnon-Buchanan cycle**.
Whereas **green non-sulfur bacteria** (*Chloroflexi*) can also live photoautotrophically by **reducing CO_2** to **glyoxylate** through the **3-hydroxypropionate bi-cycle**. Another CO_2 fixation pathway, found in obligate anaerobic methanogens, acetogens and few other **chemolithoautotrophs**, is the reductive acetyl-coenzyme A pathway, or **Wood-Ljungdahl pathway**, which, differently from the others, it's not a cycle.

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CARBON FIXATION IN PHOTOSYNTHETIC ORGANISMS

Reductive pentose phosphate cycle
(Calvin cycle)

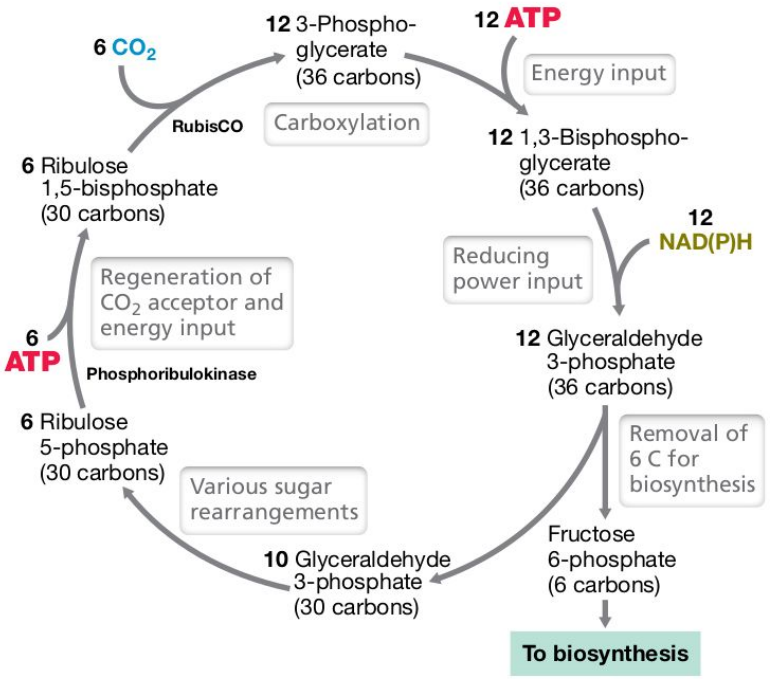


https://www.kegg.jp/pathway/map00710

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Calvin-Benson-Bassham cycle

The Calvin Cycle was discovered in 1950 in green algae. It is present in cyanobacteria, purple bacteria, algae, and many chemolithotrophs. It is a high energy-demanding pathway, requiring 3 mol ATP and 2 mol NAD(P)H per 1 mol of CO_2 entering the cycle, or 18 mol ATP and 12 mol NAD(P)H per 1 mol of glucose. **Two key enzymes** are involved in the pathway, a **ribulose bisphosphate carboxylase** (RubisCO) involved in the **carboxylation** of ribulose-1,5-bisphosphate, and a **phosphoribulokinase** that catalyzes the **ATP dependent phosphorylation** of ribulose-5-phosphate to obtain new molecule of ribulose-1,5-bisphosphate to restart the cycle. RubisCO is responsible for the vast majority of biological CO_2 uptake and thus biomass production on Earth. It represents 30-50% of the soluble proteins in a plant leaf and it has been estimated that for every person on Earth there is 5 kg of RubisCO.



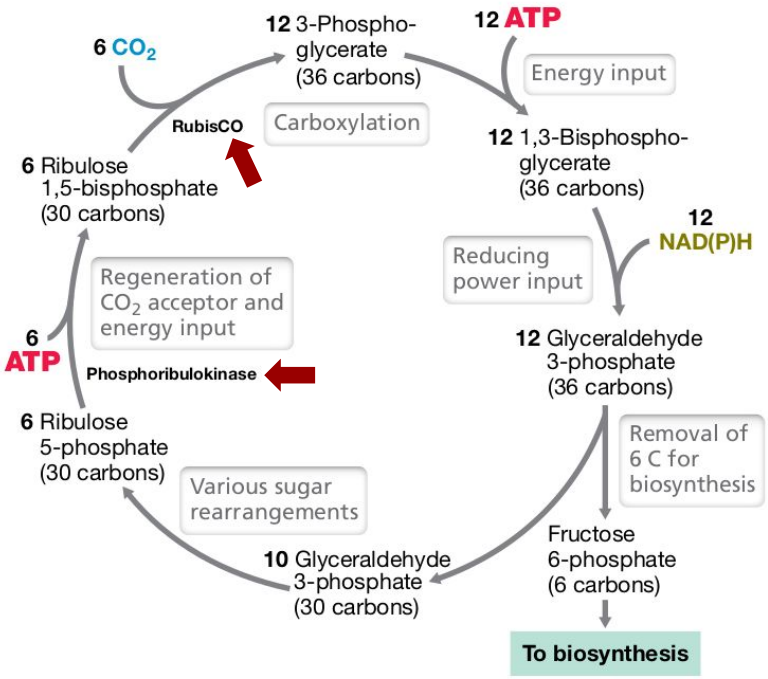
Overall stoichiometry:

$6\ CO_2 + 12\ NADPH + 18\ ATP \longrightarrow C_6H_{12}O_6(PO_3H_2) + 12\ NADP^+ + 18\ ADP + 17\ P_i$

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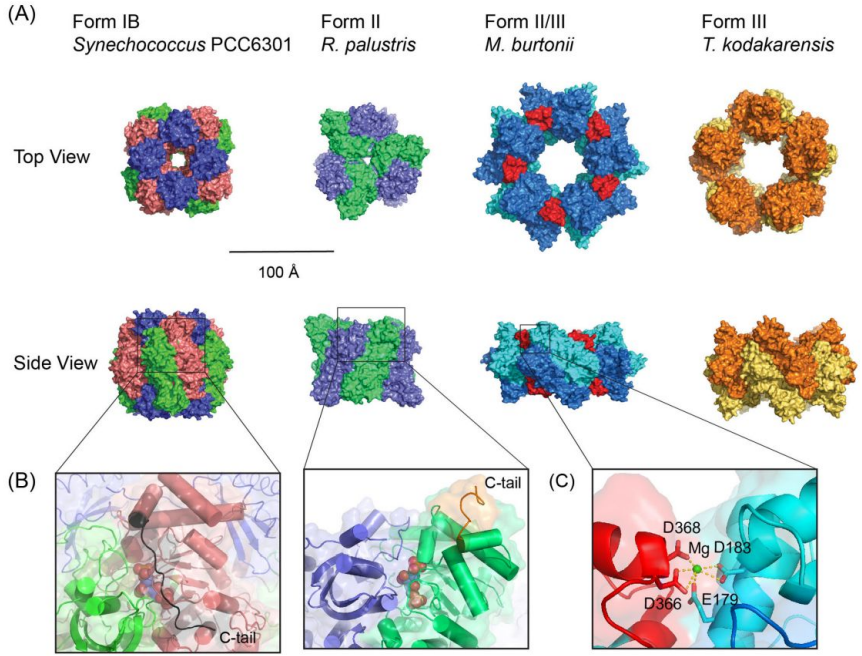
Overall stoichiometry:

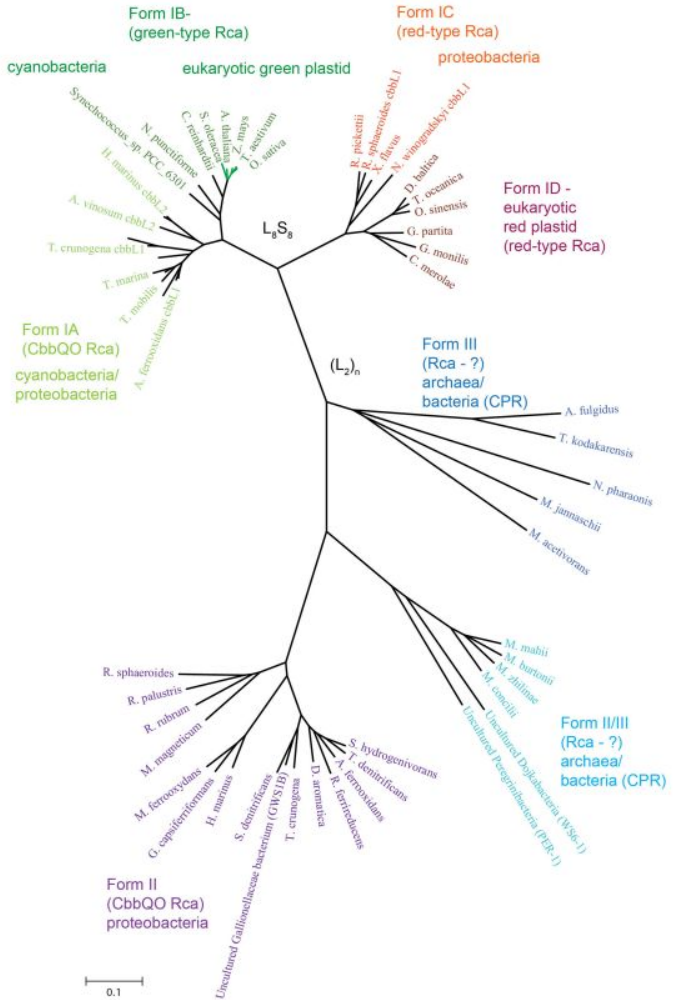
$6\ CO_2 + 12\ NADPH + 18\ ATP + 12\ NADP^+ + 18\ ADP + 17\ P_i \longrightarrow C_6H_{12}O_6(PO_3H_2)$

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Ribulose Bisphosphate Carboxylase (RubisCO)

RuBisCo is a large (500 kDa) heterodimeric enzymes formed by large and small subunits. Different forms of RuBisCo enzymes are known and widely distributed on the Tree of Life

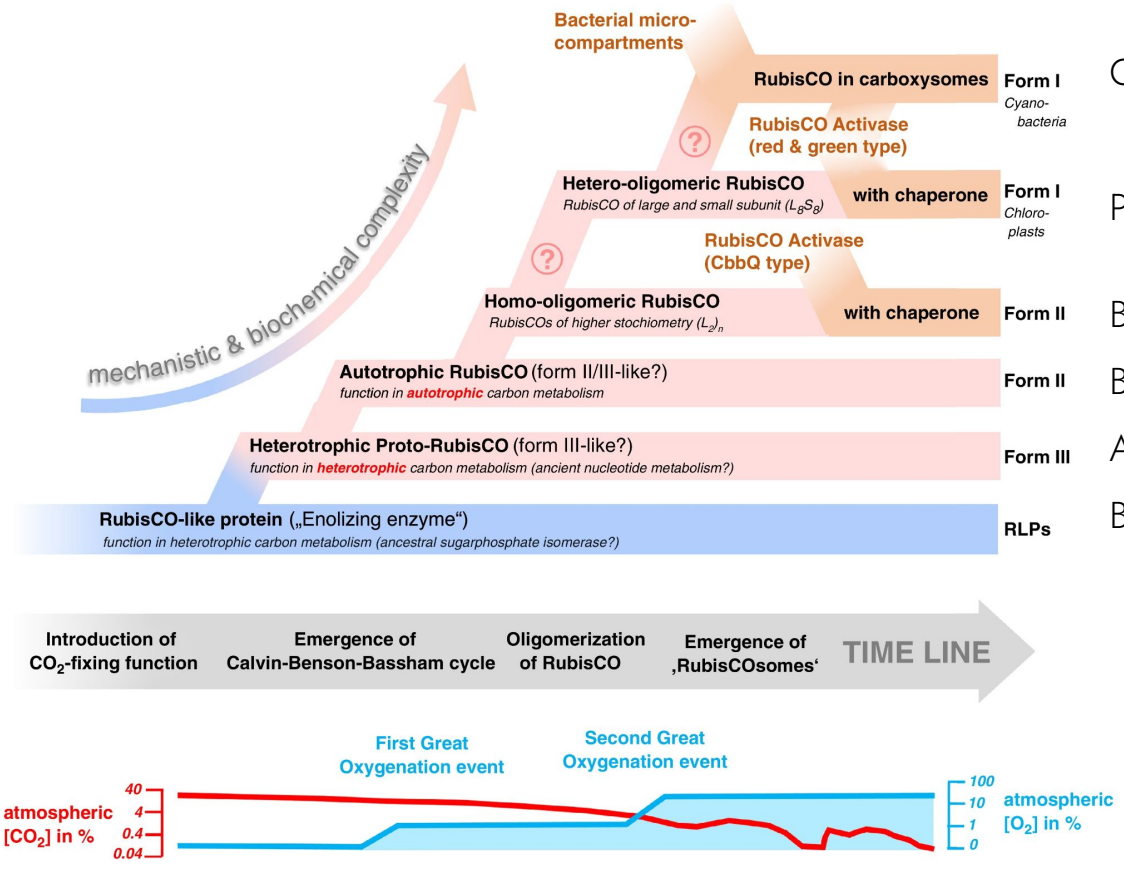




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10.1093/femsle/fnx156

Calvin-Benson-Bassham cycle



Cyanobacteria

Plants and Algae

Bacteria, chemolitotrophs

Bacteria, chemolitotrophs

Archaea, unclear function

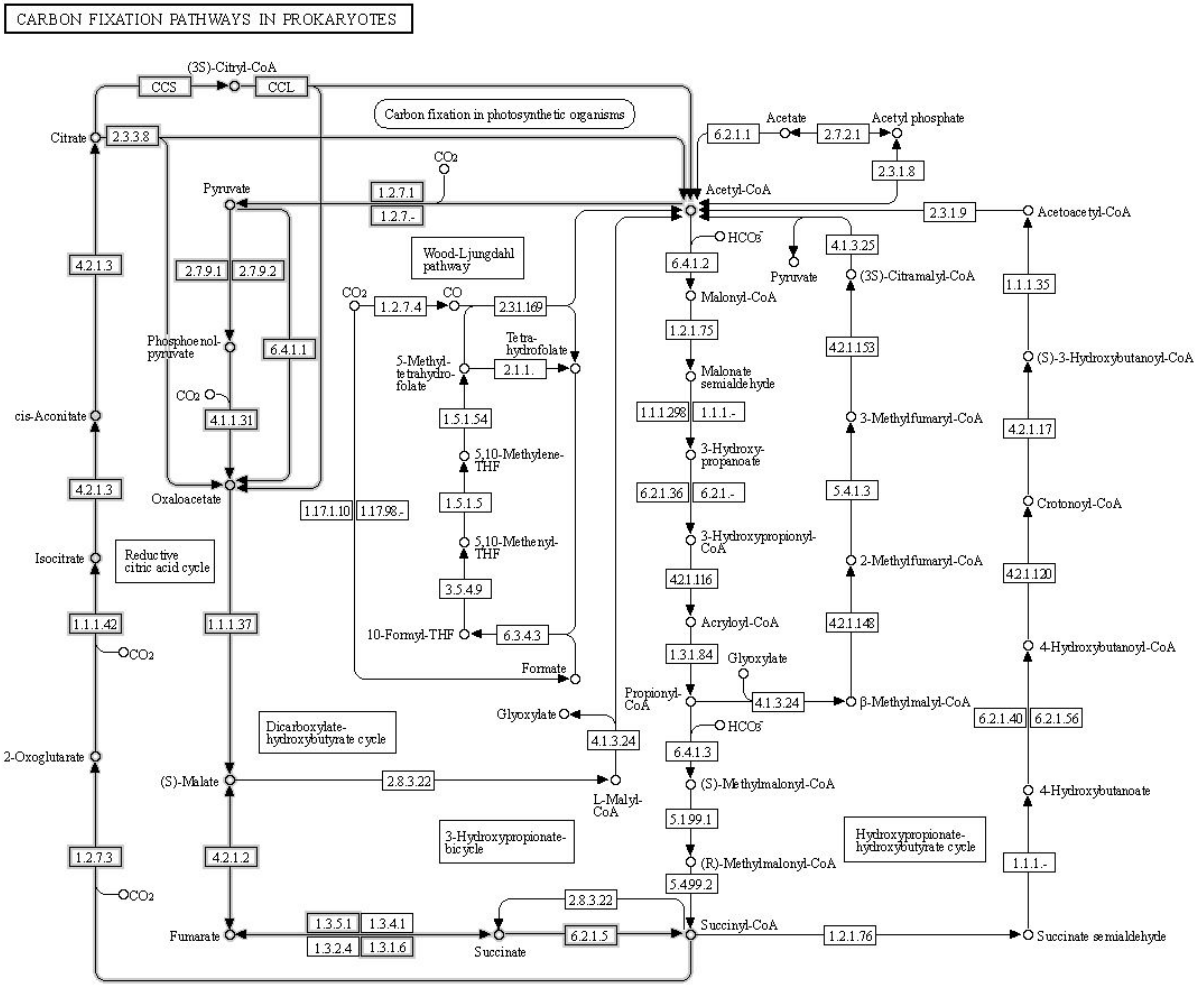
Bacteria and Archaea, unclear functio

Hypothetical timeline of the emergence and evolution of RubisCO from a non-CO_2-fixing ancestor to complex, modern-day RubisCO

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10.1016/j.copbio.2017.07.017

Reductive citrate cycle
(Arnon-Buchanan cycle)

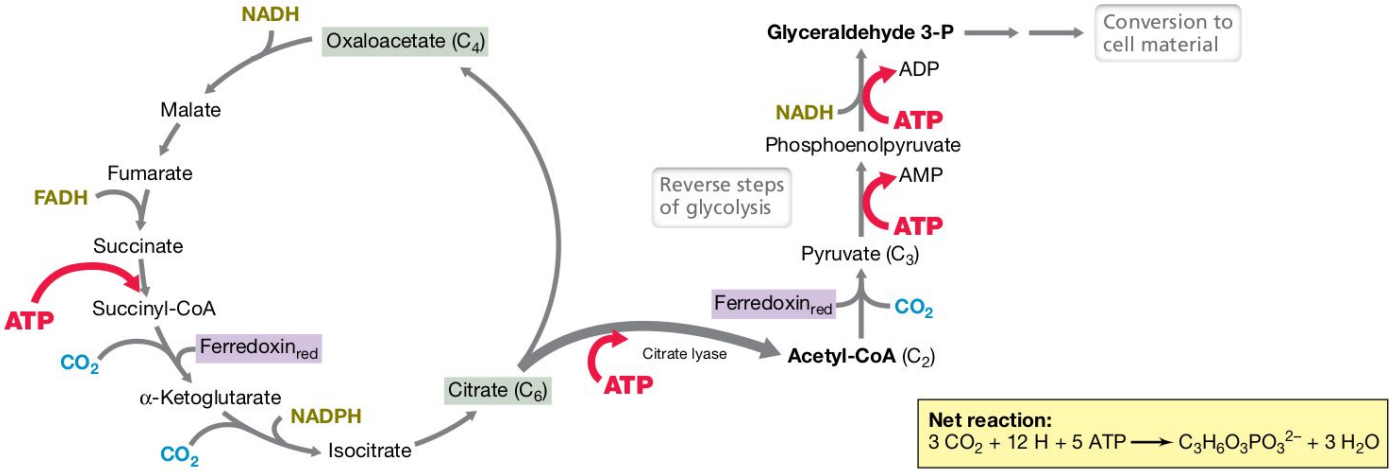


https://www.kegg.jp/pathway/map00720

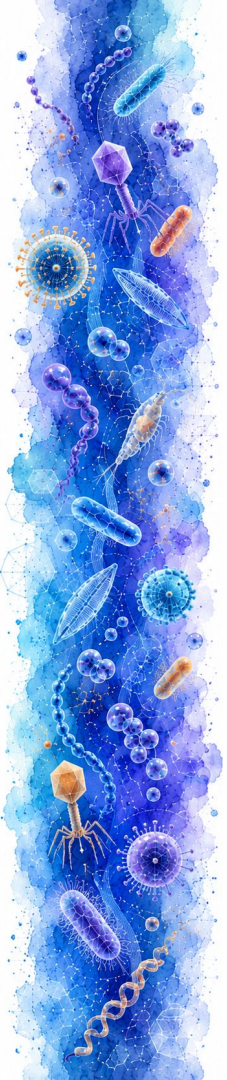
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Arnon-Buchanan Cycle

Also known as the **reductive TCA cycle**, was discovered in 1966 in in photosynthetic green sulfur bacteria (Chlorobi), it was the first of the new carbon fixation pathways to be identified. It is abundant in numerous anaerobic chemolithoautotrophic bacteria and archaea and some anoxygenic photoautotrophs. It is a more efficient pathway than the Calvin cycle, only requiring 10 ATP to synthesize 1 glucose from $6CO_2$. ATP-citrate lyase, Succinyl-CoA carboxylase and 2-Oxoglutarate synthase are the key enzymes for its functioning in reverse and are extremely oxygen sensitive. As reducing power in this cycle are used both NADPH and Ferredoxin, a low potential electron carrier acting as redox cofactor.



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Arnon-Buchanan cycle

Acceptance of new discoveries is sometime very slow.

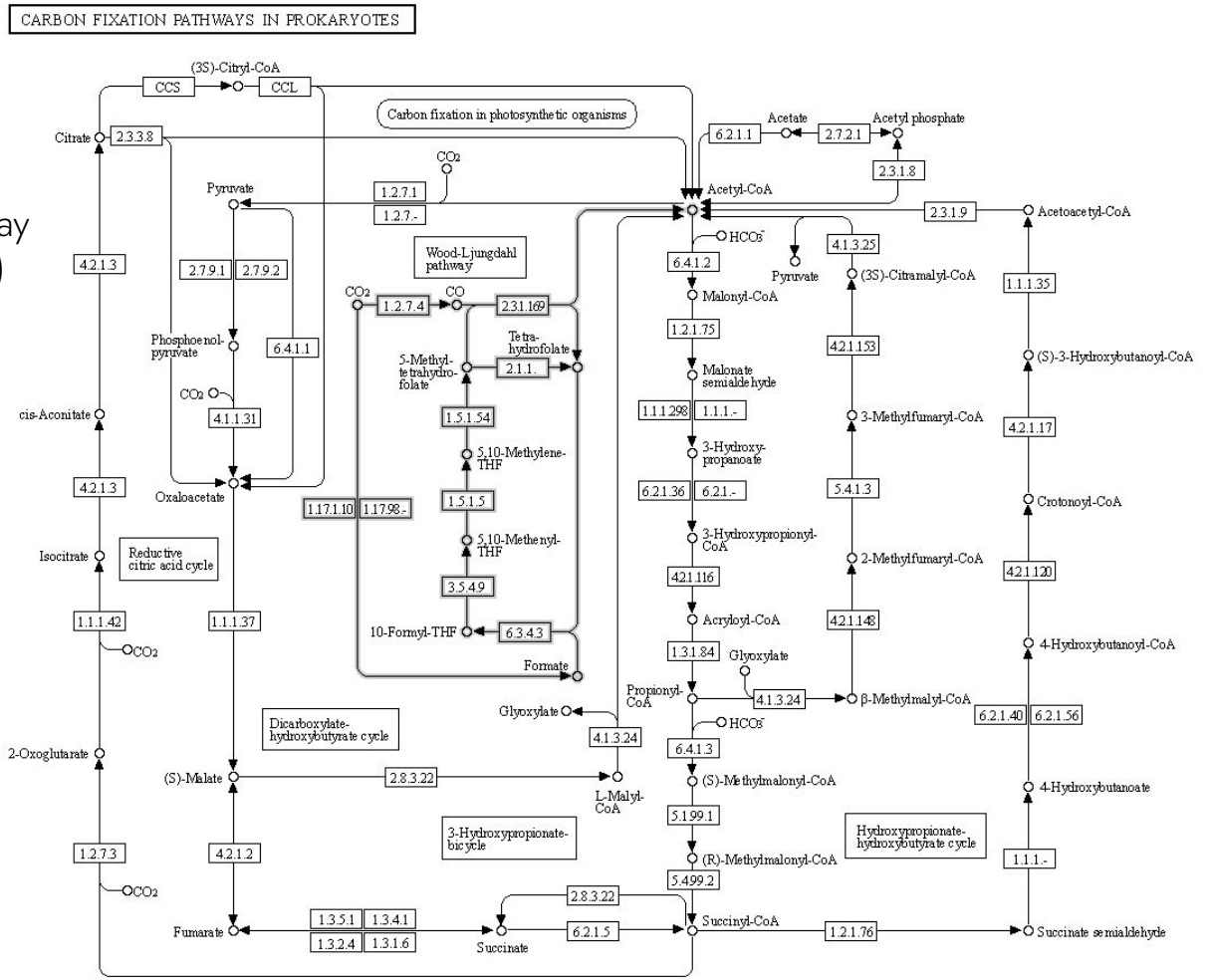
Arnon and co-workers discovered that some organisms living in reducing environments run a version of the well-known TCA cycle in the reductive direction, to the astonishment of many in the biochemistry community. At that time the Calvin Cycle was considered the only autotrophic pathway.

"Looking back, I believe the widely held view that autotrophy was synonymous with Rubisco was the major barrier to acceptance of the cycle by the scientific community. Once confirmed, the Chlorobium work was the death knell for this long-standing dogma. Thus, some ***25 years after*** *our publication in the Proceedings of the National Academy of Sciences (Evans et al. 1966) the Arnon–Buchanan cycle began to appear in textbooks of microbiology and related fields (for example, see Madigan et al. 1996)."*

Buchan
an

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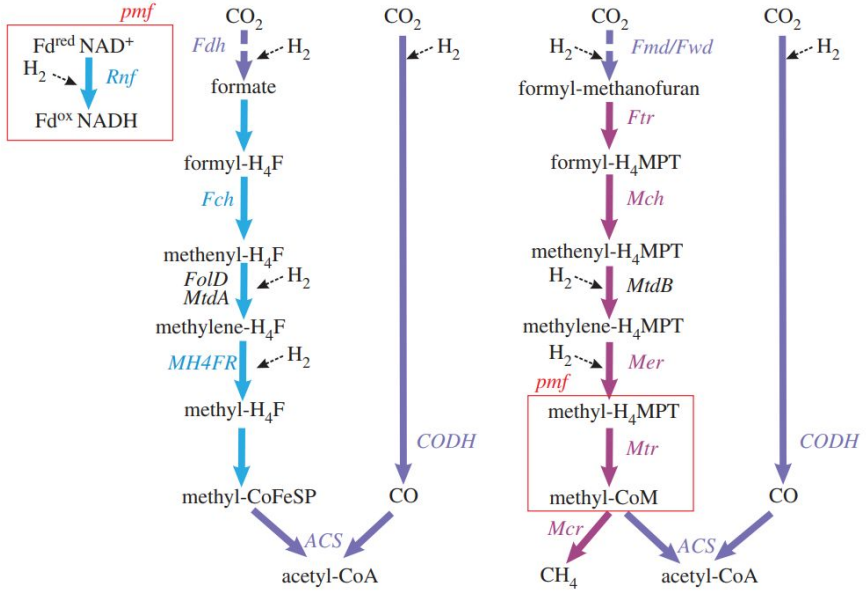
Reductive acetyl-CoA pathway
(Wood-Ljungdahl pathway)



https://www.kegg.jp/pathway/map00720

Wood-Ljungdahl Pathway

The **reductive Acetyl-CoA pathway**, discovered in 1986 in **acetogenic bacteria**, and later reported in **methanogenic archaea** and in sulfate-reducing bacteria. It is a carbon fixation pathway present in several bacterial and archaeal group, and it has been **hypothesized to be the most ancient autotrophic carbon fixation pathway**.
The pathway can be divided into **two linear branches**: the **methyl branch**, where CO_2 is consecutively reduced to a cofactor-bound methyl residue, and the **carbonyl branch**, where another molecule of CO_2 is reduced to an enzyme-bound carbonyl residue. The key enzyme is a **bifunctional carbon monoxide dehydrogenase** (CODH)/acetyl-CoA synthase. **Methanogens** couple it to the **methyl coenzyme M reductase** (mcr) to produce **CH_4**

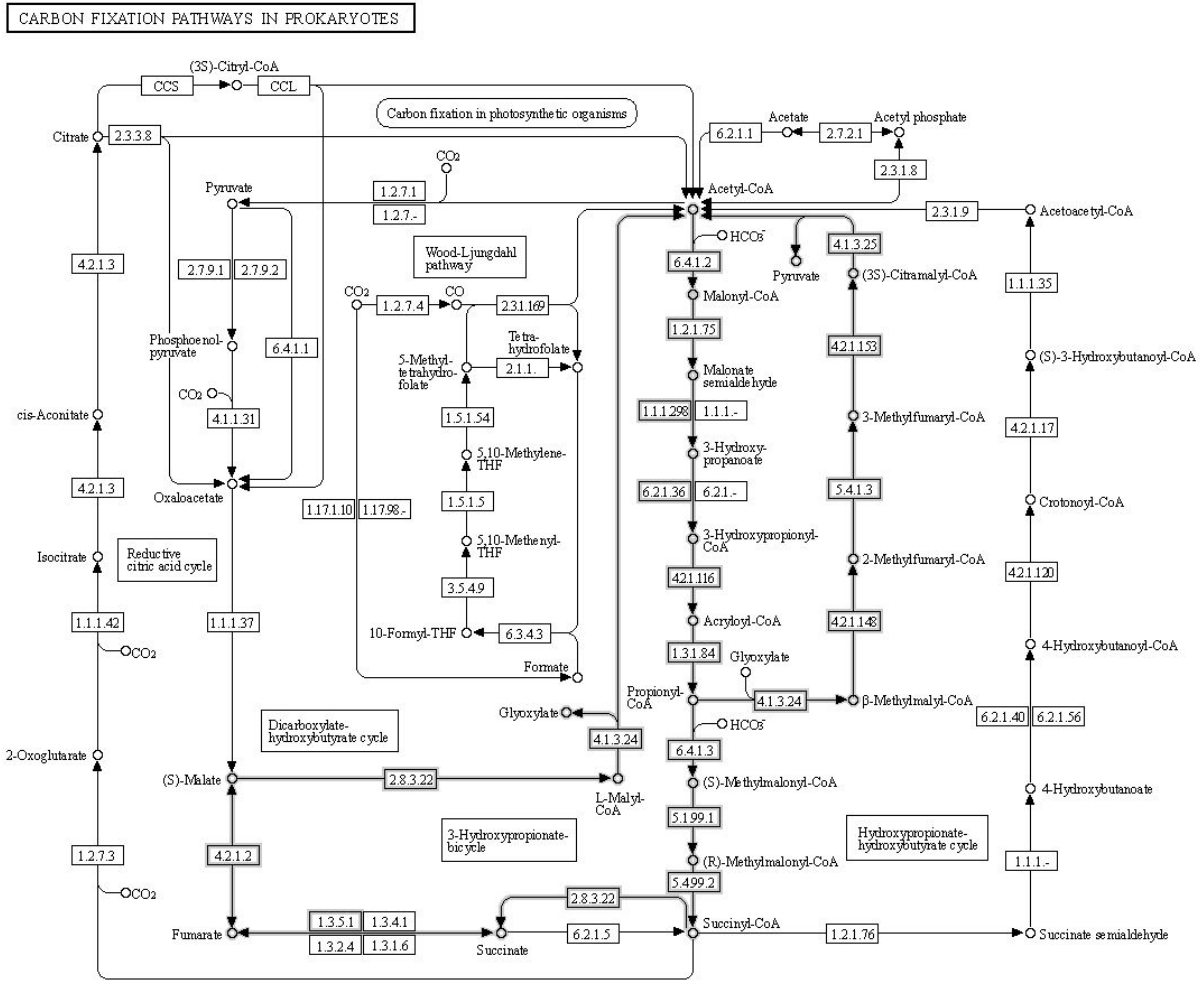


Schematic of the main reaction steps and enzymes involved in acetogenic and methanogenic WL-type pathways

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10.1098/rstb.2012.0258

3-Hydroxypropionate bi-cycle

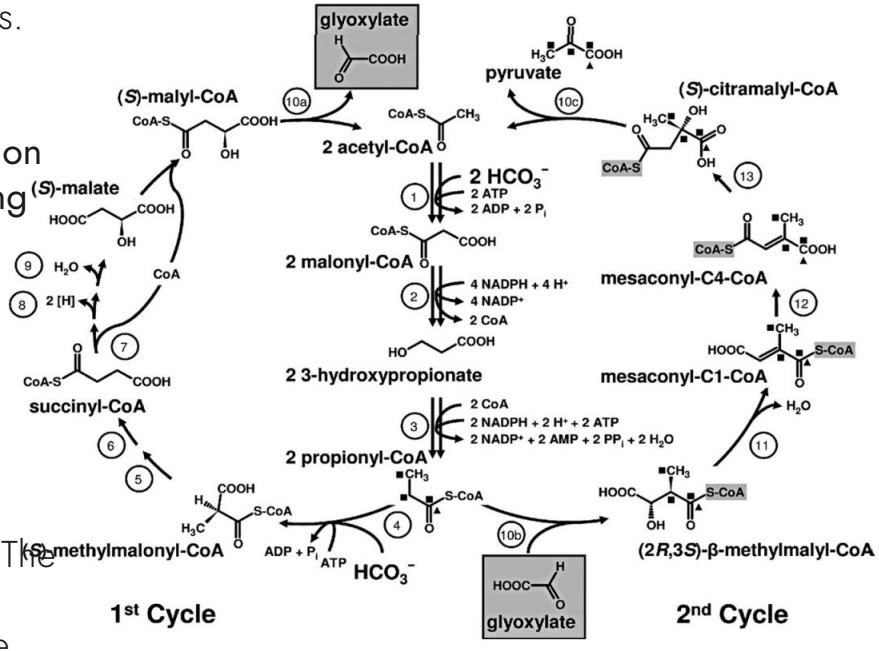


https://www.kegg.jp/pathway/map00720

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3-Hydroxypropionate bi-cycle

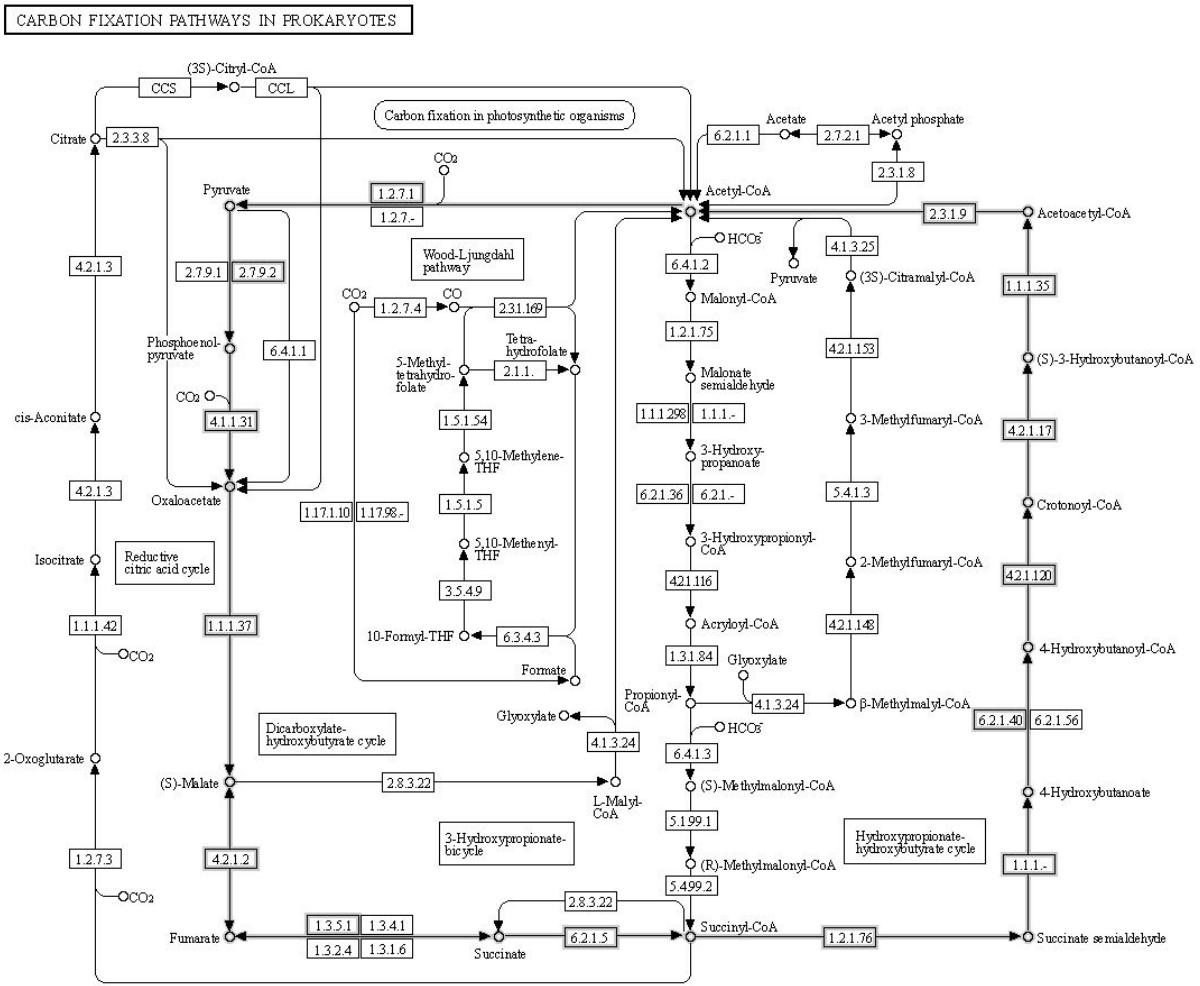
3-Hydroxypropionate bicycle was first described in the archaea *Chloroflexus antarticus* in 2002 by Herter and colleagues. It starts with an **acetyl-CoA carboxylation** by a propionyl-CoA carboxylases. In a **first cycle**, (S)-malyl-CoA is formed from 2 propionyl-CoA and 1 bicarbonate. (S)-Malyl-CoA cleavage **releases the CO_2 fixation product glyoxylate and regenerates the starting acetyl-CoA**. In a **second cycle**, glyoxylate is combined with propionyl-CoA leading to the formation of a (S)-Citramalyl-CoA, which is cleaved into acetyl-CoA and pyruvate by a tri-functional malyl-CoA lyase, which is also involved in the cleavage **(S)-malyl-CoA**. The energy costs of the 3-HP bicycle are high, it requires 7 ATP for the synthesis of 1 pyruvate. The bicycle allows **co-assimilation** of numerous fermentation products, thus being best suitable for mixotrophy and photoheterotrophy.



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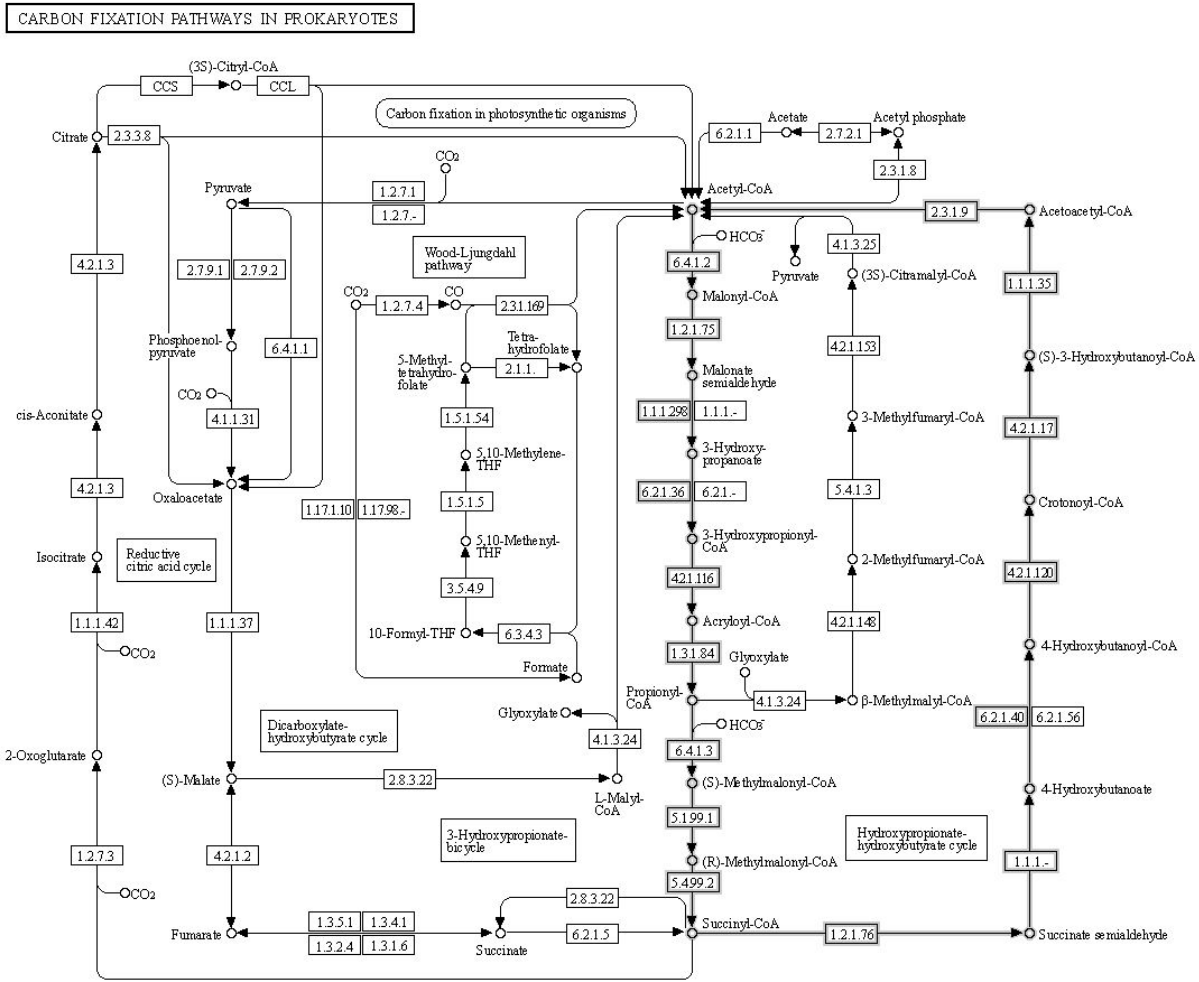
Dicarboxylate-4-hydroxybutyrate cycle



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https://www.kegg.jp/pathway/map00720

3-Hydroxypropionate-4-hydroxybutyrate cycle

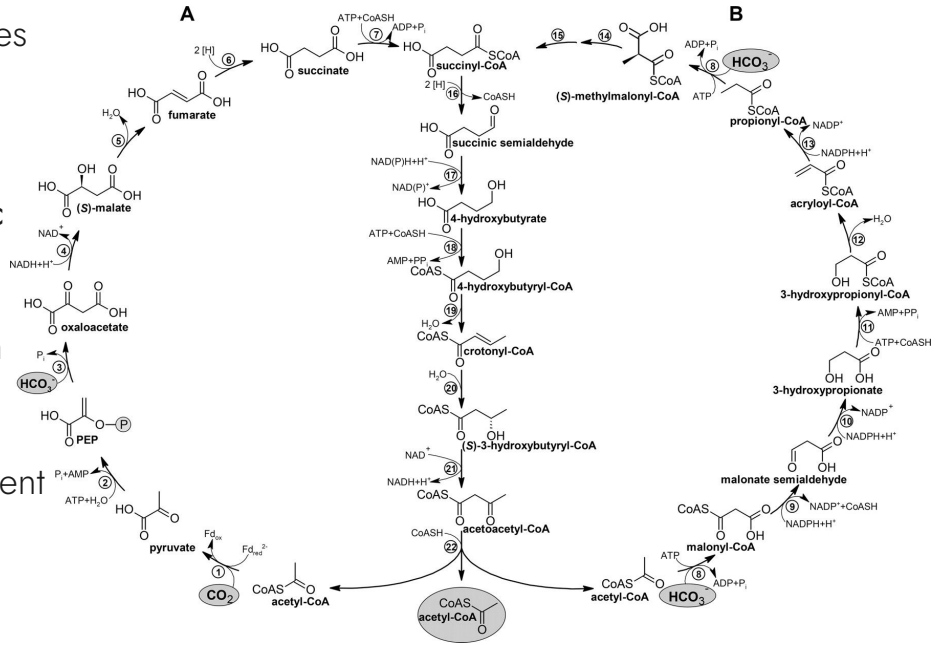


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4-Hydroxybutyrate Cycles

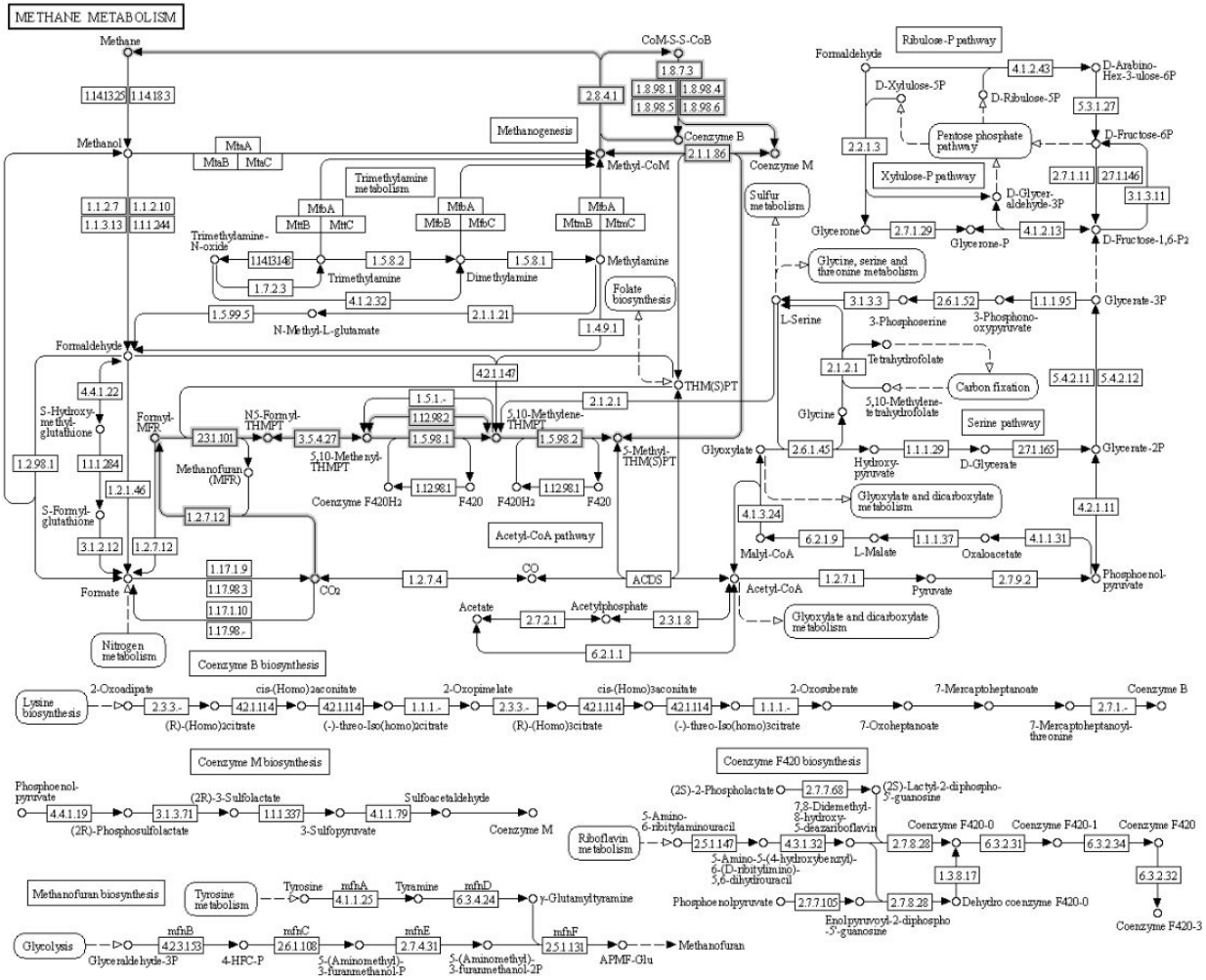
The 4-hydroxybutyrate cycles include dicarboxylate/4-hydroxybutyrate and 3-hydroxypropionate/4-hydroxybutyrate cycles that have been recently reported in Crenarchaeota, Desulfurococcales and Sulfolobales, respectively (2007/8).
In both cycles **acetyl-CoA** and **two inorganic carbons** are converted to **succinyl-CoA**, although throughout the activity of **different carboxylases**. The HP/HB cycle is present in aerobic organisms (**oxygen-tolerant**, ATP-dependent acetyl-CoA/propionyl-CoA carboxylase), while the DC/HB cycle is present in anaerobic organisms (**oxygen-sensitive** ferredoxin-dependent pyruvate synthase and phosphoenolpyruvate carboxylase).
They are both energy-demanding processes requiring **9 ATP** and **5 ATP** for the synthesis of **1 pyruvate**.



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Wolfe cycle

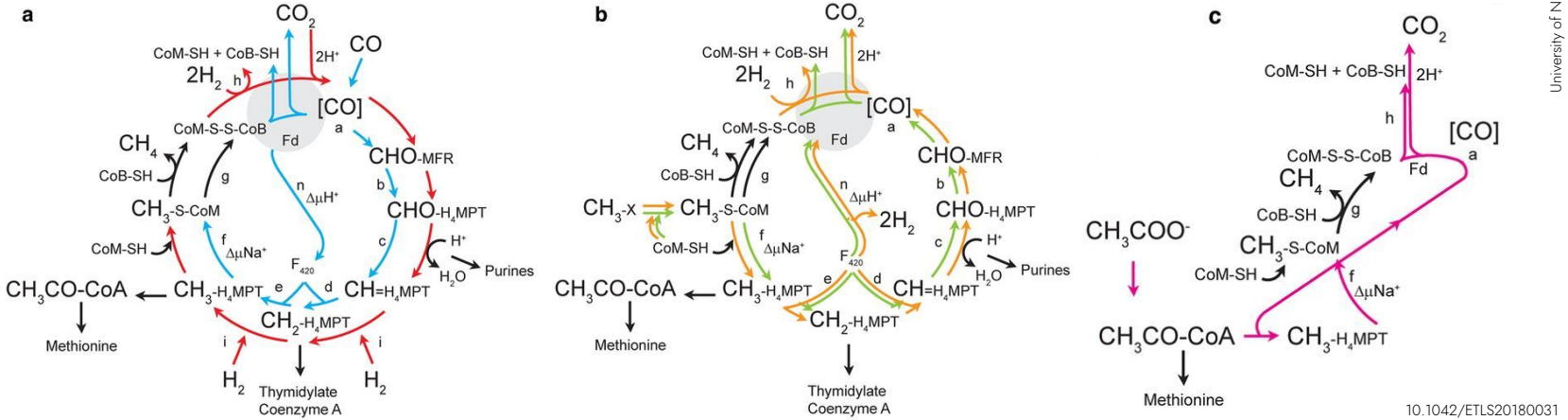


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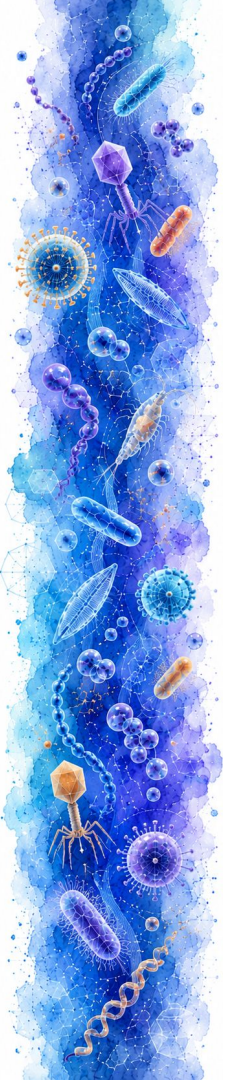
Wolfe cycle

Wolfe cycle was originally proposed in 1988, when Rouvière and Wolfe suggested that production of methane by hydrogenotrophic Archaea was carried out in a cycle rather than a pathway. Essentially, it is a **cyclization of the reductive Acetyl-CoA pathway** (Wood-Ljungdahl pathway). The nature of the cycle was confirmed only in 2012 by Lie and colleagues. Wolfe Cycle reduces CO_2 to CH_4 with H_2 in hydrogenotrophic (a), methyl-compounds in methylotrophic (b), and acetate in aceticlastic methanogens (c). The cycle starts with the **reduction of CO_2 to formylmethanofuran** (CHO-MFR) (or acetyl-coA). In the last step, the reduction of the heterodisulfide (CoM-**S-S**-CoB) to coenzyme M (CoM-SH) and coenzyme B (CoB-SH) is catalyzed by the hydrogenase EhaA-T, and is coupled with the reduction of a ferredoxin required for CO_2 reduction.



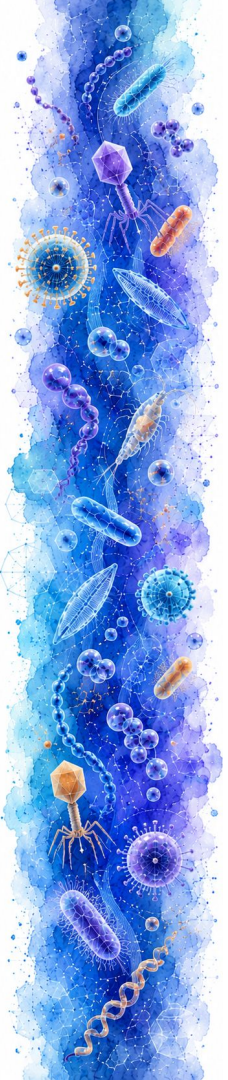
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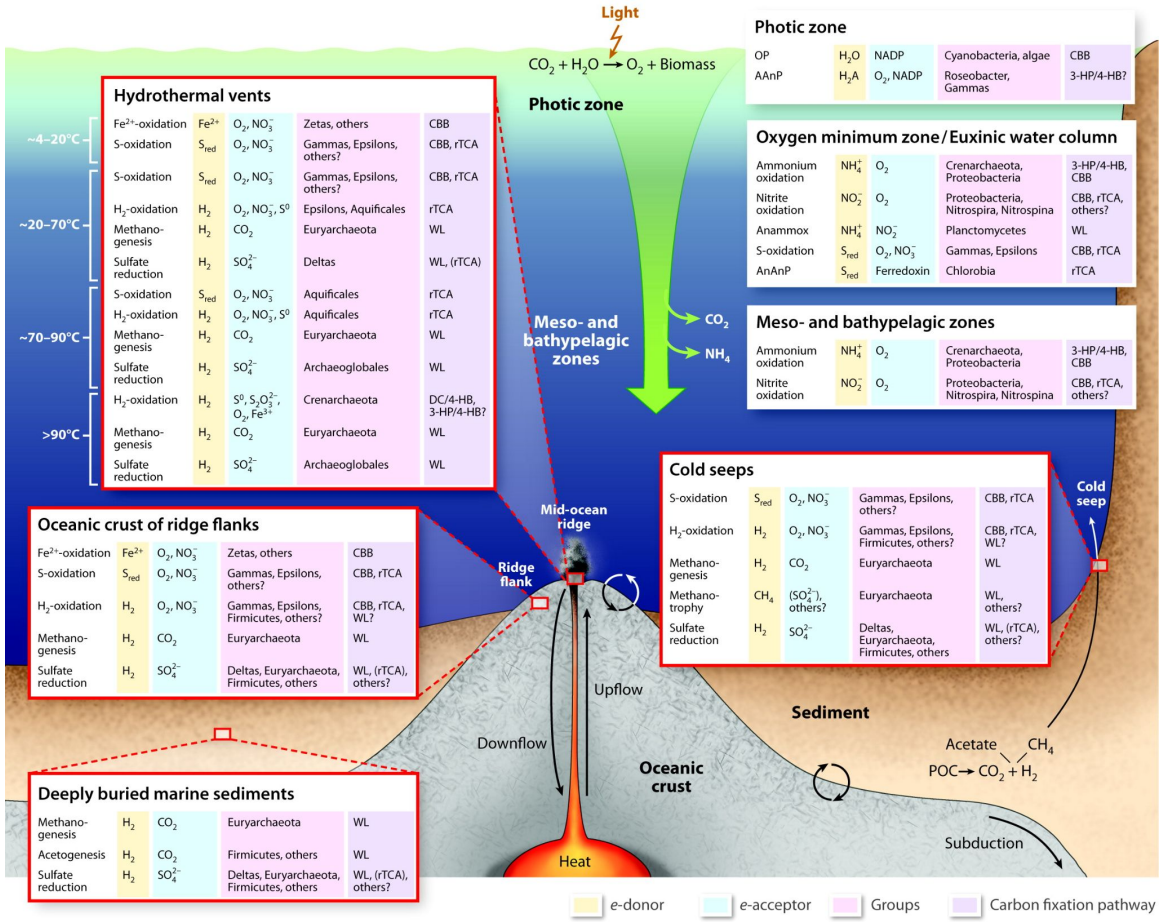
Pathway	Distribution
Calvin Cycle	Plants, algae, *Cyanobacteria*, most aerobic or facultative aerobic chemolithoautotrophic *Bacteria*
rTCA Cycle	*Chlorobiales*, *Aquificae*, *Epsilonproteobacteria*, some *Deltaproteobacteria*, few *Alphaproteobacteria* (Magnetococcus), one *Gammaproteobacteria*, *Nitrospirae* (anoxygenic phototrophs and both chemolithotrophic bacteria and archaea)
rAcetyl-CoA	Methanogenic and sulfate reducing *Euryarchaeota*, acetogenic *Firmicutes*, some *Spirochaetes*, many *Deltaproteobacteria*, Annamox bacteria of *Plancomycetes* (anaerobic bacteria and archaea)
3-hydroxyproprionate Bycycle	*Chloroflexaceae* (green non-sulfur bacteria)
DC/4HB Cycle	Anaerobic *Thermoproteales*, *Desulfurococcales* (*Crenarchaeota*)
3HP/4HB Cycle	Aerobic *Sulfolobales* (*Crenarchaeota*), marine and soil ammonia oxidizing *Thaumarchaeota*

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Pathway	ATP for 1 Pyruvate	Environment	Key enzymes
Calvin Cycle	7	Aerobic and Microaerophilic	RubisCO; phosphoribulokinase
rTCA Cycle	2	Anaerobic and Microaerophilic	2-Oxoglutarate synthase; Succinyl-CoA synthase; ATP-citrate lyase
rAcetyl-CoA	1	Anaerobic	Formate dehydrogenase; F-MTF synthase; Acetyl-CoA synthase/ CO dehydrogenase
3-hydroxyproprionate Bycycle	4	Anaerobic	Malonyl-CoA reductase; propionyl-CoA synthase; malyl-CoA lyase
DC/4HB Cycle	5	Anaerobic	4-Hydroxybutyryl-CoA dehydratase
3HP/4HB Cycle	9	Aerobic	Acetyl-CoA/propionyl-CoA carboxylase

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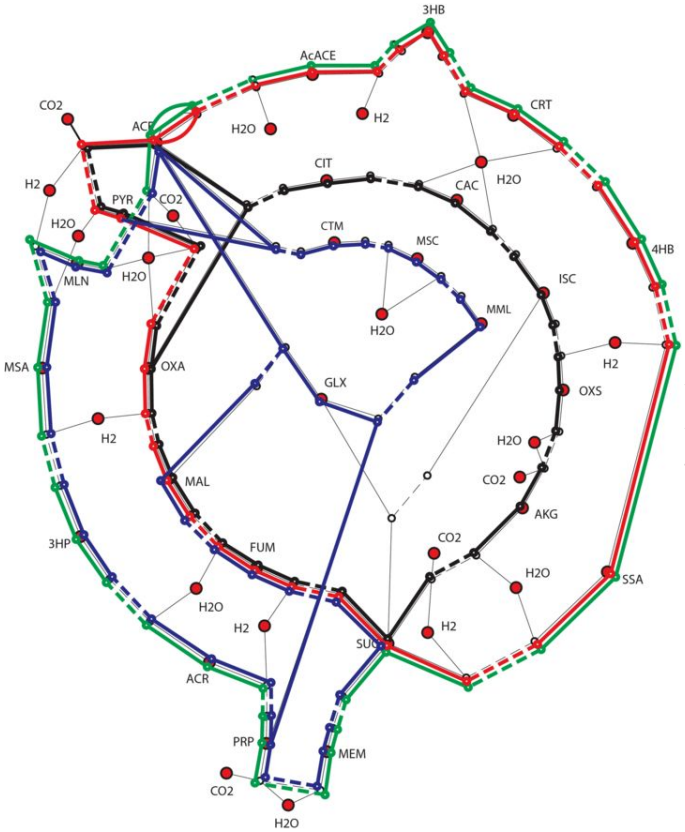
Alternative carbon-fixation pathways provide important contributions to carbon cycle in many oceanic environments, most notably deep-sea hydrothermal vents, cold seeps, meso- and bathypelagic ocean, oxygen deficiency zones, redoxclines, and euxinic waters. Of particular relevance are the **rTCA cycle**, which appears to be the main carbon fixation pathway at deep-sea vents, the **3-HP/4-HB** cycle most likely operating in nitrifying crenarchaeota in the dark pelagic realm, and the **WL-pathway** operating in methanogenic archaea and potentially in anammox planctomycetes.

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Carbon Fixation and Central Metabolites

Despite their diversity, known **carbon fixation pathway** (and central carbon pathways) **share a large number of intermediate metabolites, cofactors and to a certain extent enzymes**. It is possible to identify modules, composed either of subsets of chemicals and reactions, subsets of functions, with a conserved internal structure

Module boundaries are often associated with the most complex reaction mechanisms, catalyzed by **highly conserved enzymes**. Cofactors form a biosynthetically and functionally distinctive control layer over the small-molecule substrate, with the most complex cofactors often associated with the reactions at module boundaries in the substrate networks. This suggests that **early evolution of core carbon-fixation, appears to have required very few innovations** to produce adaptations to simple chemical or energetic differences of environment without diverse solutions and without historical contingency.



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10.1088/1478-3975/10/1/011001

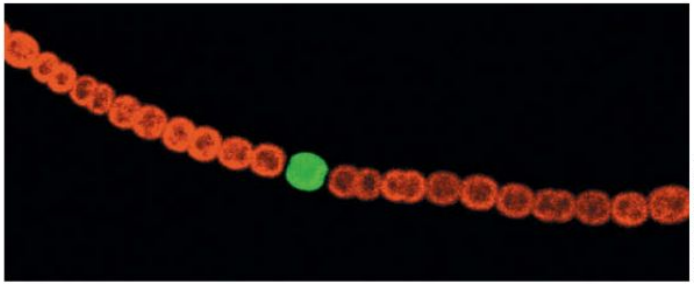
N_2 fixation

In addition to carbon, cells need a significant amount of nitrogen to synthesize proteins, nucleic acids, and many others. **Nitrogen fixation** is the key pathway allowing for the **transformation of gaseous N_2 to forms that can be used by a wide diversity of other organisms**. It is not widespread among microorganisms and the key enzyme is nitrogenase. Nitrogenase genes have been found in 9 bacterial phyla and 1 archaeal phylum. Nitrogen-fixers are known as **diazotrophs** and can be **both free living and symbionts**, forming symbiotic relationships with plants, animals, and fungi.

Nitrogen fixation is **inhibited by oxygen** because nitrogenase is irreversibly inactivated by O_2. In some nitrogen-fixing cyanobacteria, nitrogenase are located into differentiated cell called **heterocysts**, where conditions are anoxic, while in neighboring vegetative cells oxygenic photosynthesis still occur.

TABLE 14.1 Some nitrogen-fixing organisms[a]

Free-living aerobes		
Chemoorganotrophs	***Phototrophs***	***Chemolithotrophs***
Azotobacter *Azomonas* *Azospirillum* *Klebsiella*[b] *Methylomonas*	Cyanobacteria (e.g., *Anabaena, Nostoc, Gloeothece, Aphanizomenon*)	*Alcaligenes* *Acidithiobacillus*
Free-living anaerobes		
Chemoorganotrophs	***Phototrophs***	***Chemolithotrophs***[c]
Clostridium *Desulfotomaculum*	Purple bacteria (e.g., *Chromatium, Methanococcus, Rhodobacter*) Green sulfur bacteria (e.g., *Chlorobium*) Heliobacteria	*Methanosarcina* *Methanocaldococcus*
Symbiotic		
With leguminous plants	***With nonleguminous plants***	
Soybeans, peas, clover, etc. with *Rhizobium, Bradyrhizobium, Sinorhizobium*	Alder, bayberry, autumn olive, many other bushy plants, with the actinomycete *Frankia*	

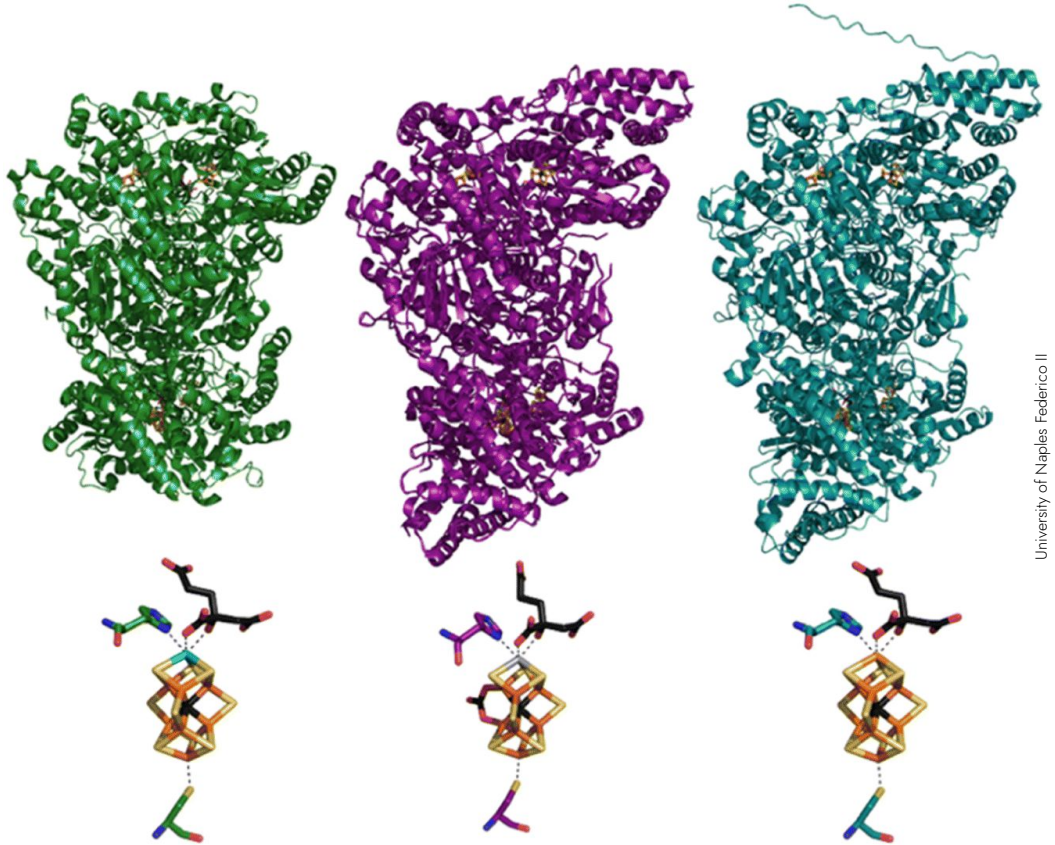


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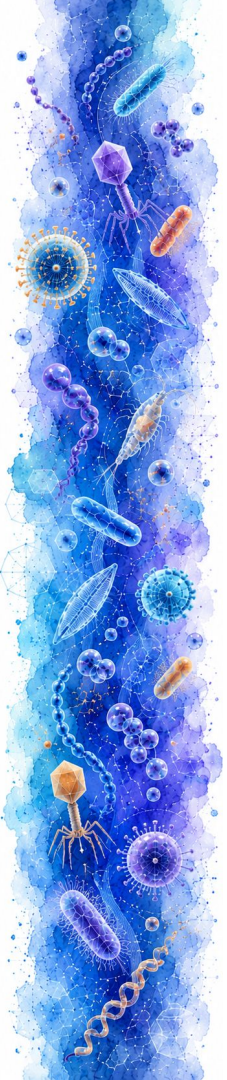
N_2 fixation

Nitrogenase is a two-component system, comprised of a **reducing Fe-protein**, and a **catalytic cofactor-containing metal-ion protein**. Three types of nitrogenases have been reported, each named for the metal contained in the catalytic cofactor: **Mo**, **V**, and **Fe**. The catalytic clusters are referred to as FeMoco (green), FeVco (purple) and FeFeco (blue). **Nitrogenase distribution** is often influenced by the availability of nitrogen in the environment. Microbes in **nitrogen-poor environments** are more likely to have nitrogen-fixing capabilities, as nitrogen fixation is energetically expensive.



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10.1039/D2FD00174H



This Week Read

Hügler M, Sievert SM. Beyond the Calvin cycle: autotrophic carbon fixation in the ocean. Ann Rev Mar Sci. 2011;3:261-89. doi: 10.1146/annurev-marine-120709-142712.

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